Recorded metadata predicts the phenotype label across public single-cell cohorts and changes what internal validation can establish

Short title: Recorded metadata and single-cell classifier validation

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Abstract

Patient-level single-cell classifiers routinely report cross-validated AUC above 0.95. A high AUC supports disease-related discrimination only if the metadata recorded about recruitment, processing and sequencing do not themselves predict the phenotype label. We measured this association in nine phenotype contrasts from five public blood datasets (809 donors), covering lupus, COVID-19, influenza, cytomegalovirus serostatus and sepsis versus COVID-19. Recorded metadata were split into collection variables (study design), demographics (case mix or risk factors) and sample-quality summaries (assay or disease state). Each group's out-of-fold AUC was tested against a permutation null built from that comparison's own metadata. Expression classifiers were then tested with a free label permutation, asking whether expression is associated with the phenotype at all, and a collection-stratified permutation, asking whether discrimination exceeds the recorded collection strata.

Metadata alone predicted the phenotype label in eight of nine comparisons, in all five seeds and after Benjamini-Hochberg correction (out-of-fold AUC 0.404 to 1.000); collection variables alone did so in four of the seven comparisons that record them. Similar overall metadata AUCs masked different sources of predictability: sample quality in one COVID-19 cohort, recruiting site in another. The comparisons fell into three validation regimes. A prespecified cytomegalovirus negative control showed no detectable metadata association. Seven comparisons were associated with metadata but had label-mixed strata, so the conditional question could be asked; in sepsis versus COVID-19 only the free test rejected. In influenza no collection stratum held both labels, so it could not be answered with these donors. Adding expression to metadata raised AUC by 0.087 to 0.354 in seven comparisons and by nothing measurable in the other two.

Metadata-phenotype association is therefore a measurable property of a benchmark cohort that changes what internal validation can support. These analyses quantify association with recorded variables; they do not establish that discrimination is biological.

Keywords: patient-level classification; single-cell RNA sequencing; study metadata; confounding; permutation tests; external validation; systemic lupus erythematosus; COVID-19

Author summary

Single-cell sequencing of a blood sample can be turned into a diagnostic score, and published scores often separate patients from healthy donors almost perfectly. Before trusting such a score, one needs to know whether it reflects the disease or the way the study was run. In many public datasets, patients and healthy donors were recruited at different hospitals, sequenced at different times or processed in different batches, and a model can score well by recognising where a sample came from.

We measured how often this happens in nine comparisons from five public datasets covering lupus, COVID-19, influenza and cytomegalovirus infection. Using only the recorded study details, without any gene expression, we could predict which group a donor belonged to in eight of the nine. The ninth, chosen in advance as a negative control, showed no such pattern. Cohorts that looked equally affected were affected through different details.

The usual validation test shuffles patient labels and asks whether expression predicts them at all. Shuffling labels only among donors collected the same way asks whether expression predicts them beyond the collection process. The two tests can disagree, and in one cohort the second cannot be run.

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1. Introduction

Single-cell RNA sequencing supports patient-level classification by capturing differences in immune-cell composition and molecular state, and published classifiers routinely separate patients from controls at cross-validated AUC above 0.95. For such a result to inform clinical research, the discrimination has to extend beyond the recruitment and processing conditions of the development cohort. This is hard to assess when the phenotype label is associated with collection site, sequencing batch or sample quality.

Many methods now produce one representation per patient: pseudobulk aggregation, phenotype-oriented summaries, prototype networks, sparse gene panels, multicellular factor models, graph representations, and methods that first identify disease-relevant cell populations [6-11,32-35]. Frozen single-cell foundation models offer another route - encode each cell, average per donor, train a small classifier - although benchmarks report that such embeddings can retain batch structure and can underperform simple expression baselines [13,14]. These methods are compared on public cohorts, and the comparison is informative only if those cohorts do not let a model succeed by recognising how samples were collected.

In public data, cases and controls are often collected differently: they come from different clinics, years, chemistries or processing batches. Batch-label confounding has been known to bias cross-validation for a decade [16], and a fully confounded single-cell design is not identifiable [17]. Regression on technical covariates does not resolve the problem, because it removes whatever part of the data the covariates predict, technical or biological [18,19]. Work in predictive modelling has shown that adjustment must be fitted inside training folds, that confound regression can leak information to nonlinear learners, and that residual confounding can survive apparently successful correction [27-29]. Benchmark studies raise the related question of how the datasets themselves are assembled [12,15].

Existing work provides statistical tools for detecting confounding and testing conditional association [25,29,30]. What remains poorly characterised in patient-level single-cell prediction is the benchmark cohort itself. It is not known how often recorded acquisition structure predicts the phenotype, which parts of the metadata carry that association, or whether standard validation procedures preserve or destroy it. Nor is it known when the available donors contain enough design overlap for a conditional question to be asked at all. We therefore treat metadata-phenotype association as a measurable property of the cohort rather than as a post hoc explanation for classifier performance, and examine its consequences across nine phenotype contrasts from five public single-cell datasets.

We organise the analysis around four questions that any patient-level evaluation answers, explicitly or not. **Presence:** does the recorded metadata predict the phenotype? **Source:** which group of variables carries that prediction? **Validation:** does expression discriminate beyond the acquisition structure preserved by the relevant null? **Transport:** does the discrimination survive a genuinely different collection process?

Recorded metadata are not one kind of variable. Collection variables describe study design in the strict sense. Demographics describe case mix and may also be risk factors. Sample-quality summaries are measured downstream of the assay and can also be affected by disease. We analyse the three groups separately, treat collection as primary for any statement about study design, and report headline counts under both definitions.

The study makes three contributions. The first is empirical: across diseases, we quantify phenotype predictability from recorded metadata as a cohort-level property of public single-cell benchmarks and decompose it into collection, demographic and sample-quality components. The second concerns validation: free and collection-stratified permutations test different nulls, they give different answers in real cohorts, and in a cohort without design overlap the stratified test has no informative reference distribution. The third concerns interpretation: two quantities often read as evidence of robustness have no single interpretation. The AUC lost after metadata adjustment depends on the shape of the adjustment matrix, and agreement in a second cohort can reflect a shared acquisition mechanism. Our contribution is therefore not a new classifier or a new permutation test, but an empirical characterisation of benchmark validity in patient-level single-cell prediction.

2. Study framework and evaluation measures

2.1. The setting

Figure 1 sets out the setting in the notation of graphical causal models [36,37]. A recruitment process **R** places each donor in the recorded design **D** - site, batch, calendar period, chemistry, sample-quality profile - and is also associated with the phenotype label **Y**, because cases and controls are usually found through different routes. The measured cells **X** are affected by **D** through technical effects and by **Y** through biological ones. A classifier maps X to a prediction Ŷ.

Two further nodes matter in practice. **C** denotes covariates caused by the disease, such as severity, WHO score, comorbidity, medication and time since symptom onset. They lie on the path Y → C → X and are not collection variables; including them in D would turn disease predictability into apparent design predictability, and Section 3.1 quantifies the effect. **U** denotes collection structure that was never recorded, which no analysis here can detect (Section 8). The graph was specified before the analysis and is released in `dagitty` syntax [38].

We draw the D-Y association as induced by R rather than as a directed edge, because both directions occur: a patient may be recruited at a referral centre because of their disease, and a centre's case mix may determine which diagnoses appear in a batch. The checks below measure the association, not its direction.

2.2. The primary measure and a secondary variance summary

Metadata-only AUC. For each comparison we fit a classifier on the columns of the recorded metadata matrix **D** alone and report its out-of-fold AUC. Its null distribution comes from 200 permutations of the phenotype label, each passed through the same fixed-hyperparameter pipeline. Four names are used consistently: the **metadata-only AUC** uses every recorded variable, and the **collection-only**, **demographic-only** and **sample-quality-only** AUCs use one group each. Each group has its own permutation test, and a nonlinear arm is run beside the linear one.

Residual label variance. With y the centred phenotype vector, we also report

V_D = 1 − R^2(y ~ D),

the share of phenotype variation that the metadata cannot explain in a linear model. V_D is a residual variance rather than an information measure, and it depends on how D is coded. It is computed out of fold with five-fold cross-fitting, because in-sample R^2 inflates with the ratio of columns to donors. The CMV comparison has 89 metadata columns for 108 donors; its in-sample V_D of 0.468 would rank it the third most affected of the nine, although its matched in-sample null mean is 0.496. Each V_D is compared with a null from 200 label permutations through the same five folds, giving p(V_D). Unless marked in-sample, p(V_D) refers to the cross-fitted statistic.

Why the AUC carries the test. V_D is an unpenalised linear R^2, so its cross-fitted version loses power as D widens. In the Ren COVID-19 comparison, 26 one-hot columns on 182 donors leave the cross-fitted V_D at its ceiling, whereas a penalised logistic model on the same matrix reaches AUC 0.81 to 0.85 ($p = 0.005$ in every seed). The metadata-only AUC is therefore the primary test and V_D a secondary summary, and disagreements between them are reported.

Scope. Both quantities concern the D-Y association in Figure 1. For a classifier to be driven by metadata, D must also affect X and the classifier must use that part of X. A high metadata-only AUC is therefore necessary but not sufficient evidence of metadata-driven classification, and a low value does not exclude association through unrecorded structure.

2.3. Two permutation tests ask two validation questions

The **free permutation** shuffles the phenotype label across all donors and refits the whole pipeline. Its null is exchangeability of the label across donors, so rejection indicates an association between expression and the label, whatever its origin. Because the shuffle also destroys the metadata-label association, a small p-value cannot distinguish disease signal from collection structure. Rejection also does not certify the absence of leakage: a label-dependent step performed before permutation, such as a feature list carried over from an earlier analysis, affects observed and permuted runs alike. In our pipeline every representation step is refitted inside the training fold on every permutation (Section 9.6); that property is established by the code, not by the test.

The **collection-stratified permutation** shuffles labels only within strata that share a collection configuration - batch, pool, site and, where it varies, sequencing chemistry. Each stratum keeps its class counts, and only the pairing of labels with molecular profiles is randomised. This restricted permutation scheme was introduced by Chaibub Neto and colleagues [30] and has since been used in the confounding literature [25]. Its null is exchangeability within the collection strata, so rejection indicates discrimination beyond what those strata retain.

The strata use collection variables only; sample-quality and demographic columns are not conditioned on. If a sample-quality variable remains associated with the phenotype within a stratum and also affects expression, the stratified null is false for a reason unrelated to disease; Section 4.4 quantifies this case. Conversely, a non-significant stratified test is a failure to reject under this conditioning set, not evidence that discrimination is entirely collection-driven. Conditioning on the full metadata, including continuous columns, would require a conditional randomisation scheme that we do not develop here.

The stratified test also requires at least one stratum containing both classes. When every stratum is label-pure, within-stratum shuffling leaves the labels unchanged and the reference distribution is degenerate. We then report the test as **uninformative**, with p = NA. The free permutation remains defined and is still reported.

2.4. Three validation regimes

Together, the metadata-only AUC and the stratum structure place each comparison in one of three regimes. The regimes are defined by the cohort, not by the classifier.

1. **Low recorded association.** The metadata-only AUC is not distinguishable from its null. No recorded variable undermines the unadjusted estimate, although unrecorded structure remains possible.
2. **Associated but conditionally evaluable.** The metadata predicts the phenotype, and at least one stratum holds both labels, so the stratified test can be run. Its result indicates whether discrimination exceeds what the collection strata retain.
3. **Structurally non-overlapping.** No collection stratum holds both labels. The conditional question cannot be answered with these donors by any analysis; answering it requires donors collected under overlapping conditions. Such comparisons are labelled **not estimable** in Table 1 and reported outside the main ordering.

Separately, a comparison is flagged **underpowered** when it has fewer than 40 donors or a minority class below 15. Flagged comparisons stay in the ordering because their instability is informative: Section 3.4 reports one whose collection-stratified p ranges from 0.057 to 0.614 across five seeds. These gates depend on donor counts and stratum structure, not on the seed. The fixed-hyperparameter pipeline is the inferential statistic throughout, because both permutation nulls are built from it; the tuned AUC is released beside it as a description, and their difference is not used as a gate.

Results

Section 3 describes the nine comparisons, Section 4 uses simulation to interpret the patterns they show, Section 5 asks whether classifiers in two lupus cohorts use the detected structure, and Section 6 brings the checks together.

3. Metadata-phenotype association across nine public comparisons

3.1. Datasets and recorded metadata

We assembled five public datasets [2,39-42] through the CELLxGENE Discover curation API and reduced each to one row per donor: log1p CPM pseudobulk expression, plus variables built from schema-guaranteed observation fields (`donor_id`, `disease`, `sex`, `development_stage`, `assay`, `tissue`) and cohort-specific collection variables recovered from the distributed object. The datasets were chosen to span different acquisition structures rather than sampled systematically; Section 9.1 gives the selection rationale and a retrospective count of eligible datasets.

The metadata variables fall into three groups.

- **Collection** (site, sub-study, sequencing chemistry, assay, batch, pool) describes how material was gathered and processed, and is not caused by the disease.
- **Demographic** (age, sex, ethnicity) describes case mix. An imbalance is a fact about recruitment, but age and sex are also risk factors for most of these diagnoses, and the two readings cannot be separated with these data.
- **Sample quality** (cells per donor, mean UMI, genes per cell, mitochondrial fraction) reflects

the assay, but can also reflect disease through cell composition, treatment or the state of the sample at collection.

Collection is the primary definition for any statement about study design, and the other two groups are prespecified sensitivity analyses. The union of all three is called the *recorded metadata*. Covariates caused by the disease - severity, outcome, comorbidity, medication, symptom timing, diagnosis fields, stage, grade, WHO score - are excluded from all groups (node C in Figure 1). Table S1 lists the raw and expanded columns entering every group; a machine check confirms that it reproduces the width of all 34 fitted metadata matrices (34 of 34 blocks) and that no label or label proxy enters any of them.

The exclusion matters. In a simulation where metadata and phenotype are unrelated, admitting one disease-caused covariate raises the metadata-only AUC from 0.488 to 0.793 and flags the cohort as significant in 100% of runs, against 5.5% without it (Figure S1, Table S2). Residualising on the enlarged set then removes 0.12 AUC from a cohort with no confounding.

Nine phenotype contrasts were formed (Table 1): lupus versus healthy in GSE174188 (CD4 subset and all cells), COVID-19 versus healthy in three cohorts, influenza versus healthy and sepsis versus COVID-19 in COMBAT, and CMV-positive versus CMV-negative in HIHA. The five cohorts contain 809 distinct donors (261, 196, 120, 124 and 108). Comparisons from the same cohort share donors, so the per-comparison counts in Table 1 do not sum to 809.

Comparison	Donors	Minority	Metadata AUC	p	Collection AUC	p	Expression AUC	+ over metadata	p free	p strat.	Verdict
Influenza · COMBAT	21	10	1.000	0.0050	1.000	0.0050	0.982-1.000	+0.000	0.0010	—	**not estimable**
Sepsis vs COVID-19 · COMBAT	111	11	0.996-0.998	0.0050	0.982-0.985	0.0050	0.894-0.962	-0.001 to +0.001	0.0010	0.0569-0.6144	**underpowered**
SLE · GSE174188 (CD4)	261	99	0.872-0.889	0.0050	0.762-0.774	0.0050	0.978-0.985	+0.087 to +0.101	0.0010	0.0010	estimable
COVID-19 · COMBAT	110	10	0.862-0.884	0.0050-0.0100	0.500-0.504	0.7413-0.8657	1.000	+0.116 to +0.138	0.0010	0.0010	**underpowered**
COVID-19 · Stephenson	115	29	0.845-0.882	0.0050	0.501-0.594	0.0149-0.2338	0.972-0.996	+0.104 to +0.141	0.0010	0.0010	estimable
COVID-19 · Ren	182	25	0.812-0.854	0.0050	0.729-0.768	0.0050	0.964-0.982	+0.100 to +0.164	0.0010	0.0010	estimable
SLE · GSE174188 (all cells)	261	99	0.819-0.832	0.0050	n/a	n/a	0.992-0.995	+0.158 to +0.172	0.0010	0.0010†	estimable
COVID-19 · Ren (10x 5' v2)	161	25	0.712-0.729	0.0050-0.0149	n/a	n/a	0.954-0.977	+0.227 to +0.262	0.0010	0.0010†	estimable
CMV · HIHA	108	45	0.404-0.518	0.3831-0.8806	0.365-0.489	0.5274-0.9453	0.894-0.933	+0.306 to +0.354	0.0010	0.0010	estimable

Table 1. The nine comparisons, ordered by metadata-only AUC. Generated from the screen output by `tools/build_table1.py`. Each comparison was run at five split seeds; values are ranges over the five seeds, and a single value means that all five agreed at the printed precision. "Minority" is the smaller label group. "Metadata AUC" uses all three variable groups; "Collection AUC" uses the collection block alone, and n/a means that no varying collection variable is recorded. All AUCs are cross-fitted with the prespecified fixed-hyperparameter pipeline, which is the statistic tested by each adjacent p-value and the one both permutation tests use. Tuned nested-CV AUCs, column counts, V_D and p(V_D) are given per seed in Table S3. "+ over metadata" is the change in out-of-fold AUC when expression is added to the metadata in one joint model (Section 3.5, Table S4). Each p is one-sided against a permutation null built from that comparison's own metadata matrix. The floors 0.0050 and 0.0010 are 1/201 and 1/1001: the metadata-only AUC and V_D nulls use 200 permutations and the complete-pipeline tests 1,000. A dash under "p strat." marks an uninformative stratified test, because every stratum is label-pure. A dagger marks the two comparisons that record no collection variable; their stratification uses tertiles of log cells per donor, so the test is sample-quality-stratified. Verdicts come from the screen's gates: **not estimable** when no collection stratum holds both labels (regime 3, Section 2.4), and **underpowered** when the minority group has fewer than 15 donors.

3.2. Recorded metadata predicts the phenotype label in eight of nine comparisons

Ordered by metadata-only AUC, the nine comparisons range from 0.404 to 1.000 (Figure 2A, Table 1), with no gap that would justify a threshold. Positions are stable across seeds: excluding the CMV control, whose near-chance value varies by 0.114, the widest five-seed spread is 0.042. Eight of the nine comparisons show significant metadata association in every seed. The count remains eight after Benjamini-Hochberg correction [24] within each seed, whether the family is that seed's nine all-metadata tests or all 34 of its variable-group tests. The ninth is the CMV negative control (Section 3.3).

Counting by p(V_D) alone would give six rather than eight. In COVID-19 Ren (26 columns, 182 donors) and COVID-19 COMBAT (7 columns, 110 donors), the cross-fitted V_D ranges across seeds from 0.847 to its ceiling of 1.000 and from 0.906 to 1.000. At the ceiling the matched null reaches the ceiling too, so p(V_D) is 1.000, whereas other seeds give 0.005. The metadata-only AUCs on the same matrices are stable, 0.812-0.854 and 0.862-0.884, and significant in every seed ($p = 0.005$ and 0.005-0.010).

The count is smaller under the collection-only definition. Seven comparisons record a varying collection variable, and in four of them the collection-only AUC is significant in all five seeds. Two of those four are the comparisons flagged as not estimable or underpowered. Among comparisons that are estimable and record collection variables, two of four are significant: lupus in GSE174188 CD4 (0.762-0.774) and COVID-19 in Ren (0.729-0.768). The two comparisons without collection variables, GSE174188 all cells and the single-assay Ren subset, are stratified on tertiles of log cells per donor and marked in Table 1. Much of the association detected by the full metadata is therefore carried by sample quality and case mix rather than by study design.

Similar overall association, different sources. Two COVID-19 cohorts of similar size have closely comparable metadata-only AUCs but different drivers (Figure 2B, C).

- **Stephenson** ($n = 115$; metadata-only AUC 0.845-0.882). Sample-quality variables reach 0.865-0.884 ($p = 0.005$ in every seed), whereas collection variables reach 0.501-0.594 and are not consistently significant ($p = 0.015$-0.234).
- **Ren** ($n = 182$; metadata-only AUC 0.812-0.854). Collection variables reach 0.729-0.768 ($p = 0.005$ in every seed), whereas sample quality reaches 0.590-0.632 and is not significant ($p = 0.050$-0.154).

The appropriate follow-up differs: library quality and harmonised preprocessing in Stephenson, recruiting site and sub-study composition in Ren. A single metadata AUC would treat the two cohorts as equally affected. Reporting by variable group is also our response to the dependence of V_D on how D is coded.

Incomplete metadata can hide association. With only sequencing chemistry recorded as collection metadata, the Ren collection-only AUC is 0.553. Recovering the recruiting hospital (12 centres, five of them case-only) and the sub-study (six source studies) raises it to 0.750, and raises the metadata-only AUC from 0.758 to 0.818. A reassuring result computed on sparse metadata is weak evidence.

3.3. A negative control specified in advance

We selected the CMV cohort as a negative control before computing any statistic. Its 108 donors were assayed with one chemistry and one suspension type, and CMV status is a serological result rather than a recruitment criterion, so the collection labels were expected to be nearly independent of serostatus. The cohort is not a single batch: its metadata record 36 batch and 46 pool identifiers, which cross into 46 strata, 17 of them holding one donor. Each donor contributes one row and one label.

As predicted, CMV shows the weakest metadata association. Across five seeds the metadata-only AUC is 0.404 to 0.518 (p 0.383 to 0.881). Collection variables give 0.365 to 0.489 (p 0.527 to 0.945), and sample quality 0.453 to 0.507 (p 0.478 to 0.716). The expression classifier on the same donors reaches AUC 0.894 to 0.933, with both permutation tests at the 1/1001 floor in every seed.

Demographic variables are the exception, with AUC 0.591 to 0.639 and p 0.015 to 0.085. After Benjamini-Hochberg correction across each seed's 34 variable-group tests, $q = 0.020$ in two seeds and 0.074 to 0.115 in the other three. Sex and ethnicity are established correlates of CMV seroprevalence, so this association may reflect aetiology as much as recruitment. We therefore state the negative control at the level of the collection block, where no seed shows association.

Two further points concern interpretation. Read without a null, the in-sample V_D of 0.468 would suggest strong association; its matched in-sample null mean is 0.496 (p(V_D, in-sample) 0.284 to 0.353), and the cross-fitted V_D is at its ceiling of 1.000 with $p = 1.000$. And no detected association does not mean no association. The collection block has 80 one-hot columns but a centred rank of 45, and the full block 89 columns at rank 54, so column counts overstate the number of independent directions tested. What the comparison shows is that the recorded collection labels carry no detectable information about CMV status at this sample size.

3.4. Free and stratified permutations answer different validation questions

The two permutation tests agree in seven comparisons. The two exceptions illustrate regimes 2 and 3 of Section 2.4.

Sepsis versus COVID-19: the tests disagree. In COMBAT ($n = 111$, minority class 11) the metadata-only AUC is 0.996 to 0.998. The free permutation gives p = 0.001 in every seed, whereas the collection-stratified permutation gives p = 0.057 to 0.614 and is significant in no seed (Figure 2D). Expression is associated with the phenotype, but there is no evidence of discrimination beyond the collection strata. With 11 donors in the minority class the stratified test has little power, so this is not evidence that discrimination is entirely collection-driven. The fixed-hyperparameter pipeline also scores 0.011 to 0.066 AUC above the tuned one here, the largest gap among the eight comparisons with a stratified p-value.

Influenza: the conditional question cannot be asked. The COMBAT influenza comparison has 21 donors, and metadata separates the labels exactly (metadata-only AUC 1.000, V_D 0.000). Every collection stratum is label-pure (Figure S2), so the stratified test is uninformative. The free permutation rejects at the 1/1001 floor in all five seeds, with an expression AUC of 0.982 to 1.000. Reported alone, the free test would suggest a well-validated classifier. Together, the two results show that recorded collection and phenotype cannot be separated with these donors, and that the test designed for this question has no reference distribution.

All three regimes therefore occur in public data: low recorded association (CMV), association with conditionally evaluable strata (seven comparisons), and structural non-overlap (influenza). In the third regime no choice of statistic recovers the missing comparison; only donors collected under overlapping conditions would.

3.5. What expression adds over the recorded metadata

The metadata-only and expression AUCs show whether each predicts the phenotype, but not whether expression adds information beyond the metadata. We therefore fitted a joint model. Within each training fold, expression is reduced to principal components fitted on the training donors, the metadata matrix is built from the same donors, and one classifier is fitted on both. The three models share the fixed-hyperparameter specification and the folds (Table S4).

In seven comparisons, adding expression raises the AUC by 0.087 to 0.354 across seeds, and the joint model is within 0.023 of the expression-only model. Lupus in GSE174188 CD4 gains 0.087 to 0.101; the single-assay Ren subset, which records the least collection metadata, gains most (0.227 to 0.262).

The two exceptions are the comparisons discussed in Section 3.4. In sepsis versus COVID-19, metadata alone reaches 0.996 to 0.998 and adding expression changes the AUC by -0.001 to $+0.001$, whereas metadata adds 0.036 to 0.104 to expression alone. In influenza both models are at 1.000 and the increment is zero in every seed. The increment thus agrees with the permutation results without requiring a permutation.

In CMV the joint model is 0.069 to 0.136 below expression alone. Appending 89 uninformative one-hot columns to 50 principal components reduces discrimination, the same property that makes residualisation on this matrix costly (Sections 4.5 and 5.3).

The increment is descriptive and carries no p-value; a test would have to hold the metadata association fixed during permutation, which the stratified test in Table 1 already does. Neither the ordering nor the increment shows that a particular classifier uses metadata-related structure. Section 5 examines that path directly.

4. Simulation: what these patterns can and cannot mean

The simulations address the patterns seen in the real data: whether the metadata statistic is calibrated, why residualisation and restriction disagree, when an external cohort tests confounding, and what the stratified test and residualisation loss can establish. We crossed seven data-generating regimes with seven levels of metadata-phenotype association and two outcome types, each predicted and scored on its own scale: 19,600 simulated cohorts of 200 donors and 100 features (Figure 3). No simulation parameter was fitted to the real cohorts.

4.1. Calibration of the metadata-association statistic

When design carries no phenotype information, $p(V\_D) \leq 0.05$ occurs in 0.025 to 0.065 of runs across the seven regimes, close to the nominal 0.05 (Figure 3A). The rate rises at $\rho = 0.25$ and reaches one by $\rho = 0.50$ in every regime. Cross-fitting and a per-cohort null provide this calibration; an in-sample estimator judged against an in-sample null does not.

4.2. Residualisation and restriction diverge in every regime

Residualisation regresses the data on metadata variables and analyses the residuals; restriction compares donors collected under matching conditions. In the additive linear regime the two agree when design carries no phenotype information and diverge as the association increases (Figure 3B). At $\rho = 0$ the unadjusted, residualised, restricted and external AUCs are 0.809, 0.809, 0.787 and 0.817; at $\rho = 0.98$ they are 0.879, 0.538, 0.724 and 0.891. Residualisation removes 0.34 AUC and restriction 0.16, although the biological effect is unchanged. At $\rho = 1$ restriction is undefined, because no stratum contains both labels.

The gap appears in every regime (Figure 3C). The continuous-outcome arm uses a continuous latent outcome, ridge regression and pairwise concordance, which equals AUC for a binary outcome. Restriction outperforms residualisation at $\rho = 0.98$ in all seven regimes for both outcome types, but the magnitude differs: 0.094 to 0.200 with binary outcomes and 0.013 to 0.038 with continuous ones. A simulation run only on continuous outcomes would understate the gap about five-fold.

4.3. Threshold alignment and external validation

Class imbalance changes two things at once: prevalence, and the alignment between collection boundaries and the phenotype threshold. In the `imbalanced` regime both thresholds are at the 80th percentile; in `imbalanced_offset` the phenotype threshold is at the 80th percentile and the collection boundary at the median. With matched thresholds, imbalance has little effect: the metadata-only AUC reaches 1.000 at $\rho = 1$ and V_D reaches 0.000. With offset thresholds, the metadata-only AUC saturates at 0.880 and V_D at 0.728. The apparent reassurance of a rare phenotype therefore comes from threshold alignment, not from prevalence. This matters for the two real comparisons with minority classes of 10 and 11 donors.

In the base regime, external AUC rises with confounding, from 0.817 at $\rho = 0$ to 0.904 at $\rho = 1$, tracking internal AUC (Figure 3D). When the target cohort has a different design mechanism, external AUC stays at 0.816-0.829 while internal AUC rises to 0.899. External validation therefore tests metadata-driven discrimination only to the extent that the second collection process differs from the first. A different accession does not ensure this: cohorts assembled by similar consortia, on similar platforms and through similar recruitment routes can reproduce the same bias.

4.4. A non-disease pathway outside the conditioning set makes the stratified test reject

The stratified permutation does not condition on sample-quality or demographic variables. We simulated cohorts of 200 donors across eight collection sites in which a sample-quality variable QC predicts the phenotype within sites at strength , and expression depends only on site and QC. The generating model contains no effect of the phenotype on expression. Both permutation tests were run as for the real cohorts (Figure S3, Table S5).

Because QC $\rightarrow$ Y and QC $\rightarrow$ X, conditioning on the collection strata leaves X and Y dependent whenever > 0. The stratified null is then false, and its rejections are correct rejections rather than type-I errors; only $= 0$ measures calibration. At $= 0$ the collection-stratified permutation rejects in 19 of 300 runs, 6.3% (exact 95% CI 3.9 to 9.7; binomial $p = 0.29$ against a nominal 5%), and the free permutation in 21 of 300, 7.0% (4.4 to 10.5; $p = 0.11$), with a mean AUC of 0.500.

As increases the two tests move together. At a mean within-stratum absolute correlation of 0.29 the stratified and free tests reject in 22.0% and 22.7% of runs; at 0.61, in 95.3% and 95.7%. Classifier AUC rises from 0.500 to 0.668 without any disease effect. A significant stratified result therefore shows association beyond the collection strata, not disease biology, because a recorded variable outside the conditioning set can produce it. Where sample-quality or demographic variables are themselves associated with the phenotype (Table S3), this possibility remains open.

4.5. Residualisation loss without metadata-phenotype association

Residualisation loss is often read as a measure of contamination. We generated cohorts in which the metadata matrix is independent of the phenotype, calibrated the biological effect to an unadjusted AUC near 0.90, and residualised on matrices of increasing width. The unadjusted and residualised models are the same function call with one added argument, so they share folds, feature filter, standardiser, PCA, classifier and scoring (Section 9.9). A second layout reproduces the level sizes of the real CMV batch-by-pool crossing: 46 levels for 108 donors, 17 of them holding one donor.

Loss occurs without any association. With one to eight columns it is small (mean 0.004 to 0.023). It peaks at 0.333 for 48 columns and 0.302 for 64, then falls to 0.222 at 80 and 0.159 at 92, the width closest to CMV's 89, as additional one-hot columns become redundant and ridge shrinks them together. Variation between runs is wide; at 48 columns the central 95% of runs spans 0.224 to 0.460.

The CMV-shaped layout realises 45 columns and costs **0.363** on average (central 95% 0.272 to 0.464), more than balanced blocks of any width and enough to contain the 0.326 observed in CMV (Section 5.3, Figure S4). The metadata-only AUC in these cohorts stays at 0.48 to 0.51. A loss of this size is therefore not evidence that anything was removed. The result applies to this adjustment procedure - ridge at a fixed penalty on a training-fold standardised matrix - and to this layout. Column count is not effective dimension, and other residualisers would trade loss against completeness differently.

5. Whether classifiers use metadata-related structure: two lupus cohorts

Section 3 measures the D-Y association. This section examines the D $\rightarrow$ X $\rightarrow$ $\hat{\text{Y}}$ path in two lupus cohorts with complete donor metadata, using frozen Geneformer [4] embeddings and two pseudobulk expression representations in one fold-contained pipeline. GSE285773 [3] is the independent transfer cohort. Figure 4A shows how cases and controls are distributed across the recorded design, and Figure 4B how well metadata alone predicts disease status.

5.1. All three representations encode batch

In GSE174188 the three representations classified disease at AUC 0.982, 0.977 and 0.975, and identified processing wave, one against the rest, at 0.9997, 0.999 and 0.997. In GSE135779 the disease AUCs were 0.870, 0.933 and 0.945, and the best single-batch AUCs 0.993, 0.961 and 0.961 (Figure 5A). The representations contain batch information; this alone does not show that a disease classifier uses it.

5.2. Batch predictability did not imply sensitivity to batch adjustment

In GSE135779, Geneformer embeddings predicted the most distinguishable batch at AUC 0.993 and both pseudobulk representations at 0.961, whereas batch alone predicted disease at AUC 0.499. Batch residualisation changed disease AUC by +0.002, −0.002 and −0.009 (Figures 5B, 6C), within split-to-split variation. Dependence on batch was therefore small, though not shown to be zero, and batch information in a representation did not imply batch use by the classifier.

5.3. Residualisation and restriction answer different questions

Restriction asks whether separation persists among donors collected under overlapping conditions; residualisation removes the part of the data predictable from chosen metadata. When metadata and phenotype are collinear these are different questions, and we compare them to show that they are not interchangeable.

In GSE174188, ridge residualisation on processing-wave fractions reduced the three representations from 0.982, 0.977 and 0.975 to 0.714, 0.747 and 0.735. Restriction to the one large, near-balanced wave, compared with size- and label-matched random subsets, cost 0.014 to 0.049 (Figure 6B). The paired difference between the two losses was 0.186 to 0.255 across six comparisons, with every descriptive interval above zero.

In GSE135779, removing the single all-case batch left 36 donors with both labels in every remaining batch and changed matched performance by at most 0.012. Five ways of removing the recorded variables gave five answers (Figure 6A). Ridge on the complete set reduced the representations to 0.516, 0.522 and 0.520, a random-forest residualiser to 0.748, 0.787 and 0.795, and overlap weighting and batch location adjustment retained 0.859 to 0.937. Every full-set removal cost more than the batch-only control.

When design predicts the phenotype, the part of the data predictable from design necessarily includes phenotype-related variation, so removing it removes disease signal whatever its origin. A large residualisation loss is therefore consistent both with removing a shortcut and with removing biology.

CMV shows this without any detected association: metadata-only AUC 0.512 under this pipeline, permutation p 0.383 to 0.881 across seeds, and $p(V\_D) = 1.000$. Residualising on the same variables reduces AUC from 0.925 to 0.599, a loss of 0.326 (Figure 6C). The matrix has 89 one-hot columns for 108 donors at a centred rank of 54. In simulation (Section 4.5), a phenotype-independent matrix with the same ragged layout loses 0.363 on average, which contains the observed value (Figure S4). The size of a residualisation loss therefore depends on the adjustment matrix and cannot be read as a measure of contamination.

Nor do residualised and restricted estimates bound a biological effect. They are estimates under different assumptions, neither is identified, and their disagreement is the informative result.

5.4. Cell-type composition reproduces the pattern

Summarising the same GSE174188 CD4 cells as naive, effector-memory, regulatory and unassigned fractions gave disease AUC 0.899 to 0.915, and the same four fractions predicted processing wave at 0.803 to 0.872. Wave residualisation reduced disease AUC to 0.689 to 0.709 (Figure S5A), whereas within-wave restriction against matched subsets cost 0.027 to 0.062, with every interval crossing zero (Figure S5B). A four-dimensional biological summary thus reproduces the pattern, which places it in the cohort rather than in foundation-model embeddings and is consistent with composition being a strong patient-stratification baseline [26].

5.5. Source restriction and cross-cohort discrimination

Restricting training to one processing wave did not improve transfer. Training on all 261 GSE174188 donors and testing in the independent 26-donor GSE285773 cohort gave 0.900, 0.944 and 0.969. Training on the dominant wave lowered every value and performed no better than a random subset of the same size (Figure 7). Within GSE174188, training on waves 2 and 3 and testing on wave 4 gave 0.842, 0.811 and 0.806, and the reverse direction 0.919, 0.926 and 0.821. More varied source donors transferred better than a restricted source.

In the other direction, from the 26-donor cohort to the 261-donor target under strict source-only rules, AUCs were 0.884 [0.843-0.920], 0.919 [0.884-0.946] and 0.926 [0.895-0.953]. Both pseudobulk representations outperformed Geneformer after paired DeLong testing [23] and Benjamini-Hochberg adjustment [24] ($q = 0.0151$, $q = 0.000603$). Three learned pooling methods on the same frozen embeddings never outperformed mean pooling, and every significant paired difference was negative (Figure S6, Table S6). Increasing model capacity improved fitting to the source cohort but did not improve cross-cohort transfer.

6. Integrating metadata assessment and predictive validation

The five checks below examine different arrows in Figure 1 and fail in different ways. They are not a validity standard: we have not shown that they are sufficient or minimal, or that passing them supports a causal claim. Their value is that they can disagree.

#	Check	Shows	Does **not** show
1	Metadata-only AUC by variable group, cross-fitted V__D, p(V__D) against the cohort's own null, overlap counts	Whether recorded metadata predicts the phenotype label, and which variable groups do	That the classifier uses that metadata; anything about unrecorded structure
2	Complete-pipeline permutation, both free **and** collection-stratified	Free: an expression-phenotype association of any origin. Collection-stratified: discrimination beyond what the collection strata retain	A mechanism, or the absence of leakage. The stratified test does not cover sample-quality or demographic association within strata (Section 4.4), and is uninformative when every stratum is label-pure
3	How well the representation predicts batch	That the representation encodes collection information	That the decision rule depends on it (Section 5.2)

#	Check	Shows	Does **not** show
4	Residualisation **paired with** restriction and matched donor controls	How far two non-equivalent adjustments disagree	Which adjustment is right; bounds on a biological effect
5	Source-only external target	Whether patient ranking survives a change of collection process	Calibration or clinical usefulness; little about confounding when the target was collected the same way (Section 4.3)

Checks 1 and 2 take minutes on a donor-level table. Check 5 alone addresses transport, and no combination of internal checks substitutes for it.

The four questions of Section 1 map onto these checks: presence and source onto check 1, validation onto check 2 with support from checks 3 and 4, and transport onto check 5. For a clinical reader, an internal cross-validated AUC measures discrimination under the development cohort's sampling conditions. A metadata-only model shows whether recorded collection differences could produce that discrimination. A conditioned validation - the stratified permutation, or restriction to overlapping strata - shows whether the discrimination exceeds those differences. An independent cohort collected differently tests transfer. Together, these evaluations provide stronger evidence for a candidate diagnostic classifier than an internal AUC alone.

Applied to the nine comparisons, the checks reproduce the three regimes of Section 2.4. CMV shows no detected metadata association. In both lupus comparisons and all three COVID-19 cohorts, the metadata predicts the phenotype but the stratified test still rejects, so discrimination exceeds what the strata retain. In sepsis versus COVID-19 the stratified test does not reject, and in influenza it is uninformative; for these two, no internal analysis settles the question, and the limitation lies in the cohort rather than in the classifier.

6.1. Three comparisons traced through the tree

Figure 8 traces four comparisons through the decision tree. Three are described here, and sepsis versus COVID-19 in Section 3.4. The walkthrough imports the strata, metadata matrix and classifier from the screen, so Q2 and Q4 use the same partition, columns and pipeline. Each branch reports what can be estimated and what is missing (Table S7). It does not prescribe an adjustment, which depends on the target quantity and the scientific question.

COMBAT influenza (n = 21) stops at Q0. Its stratum variable, the recording institute, gives two strata, neither containing both a case and a control (Figure S2). The stratified test is uninformative, and no conditioned estimate can be formed. This is the only stopping criterion. The metadata-only AUC of 1.000 is consistent with the same structure but does not trigger the stop, because a finite-sample AUC of 1.000 records complete separation of these donors rather than proving non-identifiability. The fixed and tuned pipelines agree closely (0.982 to 1.000 and 0.964 to 1.000), and the free permutation alone gives $p = 0.001$ in every seed.

CMV HIHA (n = 108) reaches Route A. The stratified permutation is defined - 46 batch-by-pool strata, 21 of them holding both labels - and both tests reject at the 1/1001 floor. At Q3 no metadata association is detected: metadata-only AUC 0.512 in the walkthrough pipeline and 0.404 to 0.518 in the screen, not significant in any seed. Route A reports that no collection association was detected and that the unadjusted estimate stands. It does not recommend residualisation, which here costs 0.326 AUC (Section 5.3), close to the 0.363 lost by a phenotype-independent matrix of the same shape (Section 4.5). An undetected association is not thereby shown to be negligible, but non-detection is not a reason to adjust.

COVID-19 Ren (n = 182) reaches Route C. Both tests reject, and the metadata association is strong: metadata-only AUC 0.832 in the walkthrough and 0.812 to 0.854 in the screen. Label-mixed strata exist, so Q4 asks whether restriction is feasible, which requires at least one stratum with five donors of each class and 40 donors in total, the power floor used throughout. Ren does not meet this requirement. Its only stratum large enough for cross-validation holds 18 donors; the restricted AUC there is 0.992 against 0.733 in size- and composition-matched random subsets, with a matched-subset standard deviation of 0.145. Residualising the 26-column matrix reduces AUC from 0.967 to 0.884, a loss of 0.083. Route C therefore reports that association is present, that both adjustments are available but disagree, and that neither is estimable with useful precision; donors collected under overlapping conditions are needed.

Restriction is rarely feasible at these strata. CMV has 21 label-mixed strata but none with five donors of each class, and Ren has one usable stratum of 18 donors. Route B assumes that restriction is a practical alternative to residualisation, but in real cohorts with fine strata it often is not. We do not coarsen the strata to make Route B available, because that would change the conditioning set used by the stratified test.

Discussion

7. Implications for patient-level single-cell evaluation

Patient-level single-cell validation is not solely a property of the classifier. It depends jointly on the phenotype-acquisition structure of the cohort and on which parts of that structure the validation procedure preserves. The nine comparisons bear on each of the four questions set out in Section 1.

Presence. Phenotype predictability from recorded metadata was measurable, graded and stable across seeds, with metadata-only AUCs from 0.404 to 1.000. It has to be judged against a null built from the same matrix: an in-sample R^2 ranks CMV, the least affected comparison, as the third most affected, because its metadata matrix is wide relative to its donor count. The count of eight in nine describes these comparisons, which are neither independent nor a random sample of public cohorts (Sections 8 and 9.1); it is not an estimate of prevalence.

Source. Similar overall metadata AUCs concealed different associations. In Stephenson, sample-quality variables reached 0.865 to 0.884 and collection variables 0.501 to 0.594; in Ren, collection variables reached 0.729 to 0.768 and sample quality 0.590 to 0.632. The overall AUCs differ by about 0.02, yet the follow-up differs: preprocessing and library quality in one cohort, recruiting site and sub-study in the other. A combined metadata AUC alone would hide this.

Validation. Free label permutation tests for any expression-phenotype association. It cannot test whether a classifier exploits collection structure, because it destroys that structure. The collection-stratified permutation [30] asks the conditional question, and across these cohorts the two tests gave three different pictures: agreement in seven comparisons, disagreement in sepsis versus COVID-19, and no informative reference distribution in influenza. The last case is a substantive finding rather than a technical failure. Some validation questions require design overlap, and without it no statistical method can recover the missing comparison. The stratified test has its own limit: it conditions on collection strata only, and a sample-quality variable associated with the phenotype within strata can make it reject without any disease effect (Section 4.4). A detected conditional association is therefore not evidence of disease biology. Studies that use label permutation to support a patient-level classifier should report the stratified version beside it, or state that no label-mixed stratum exists. The increment of expression over metadata points the same way: 0.087 to 0.354 in seven comparisons, and essentially zero in the two where the stratified test did not support discrimination beyond collection.

Transport. Agreement in an external cohort is weaker evidence than it appears when that cohort shares the source's acquisition mechanism; in simulation, external AUC then rises with confounding. Its value depends on how different the second collection process is, not on whether the data carry a different accession. Residualisation loss has a similar limit: it depends on the shape of the adjustment matrix, and in CMV it reached 0.326 with no detected association.

For representation research, pseudobulk outperformed frozen embeddings in strict source-only transfer in our lupus data, and learned pooling did not change this. More generally, internal accuracy in these cohorts does not measure transportable discrimination, and no model can correct a cohort in which design predicts the phenotype. That requires recording and releasing collection metadata, and building cohorts in which cases and controls are collected together within batches.

None of these results supports a diagnostic claim; the external AUCs rank 261 or 26 donors and are not calibrated probabilities. The contribution is an evaluation layer rather than another representation learner, and it addresses questions a model leaderboard cannot. We suggest that patient-level classifiers be reported with three additions: the metadata-only AUC of the development cohort by variable group; both permutation tests, or a statement that the stratified test is uninformative; and an external result on a cohort collected through a different route, fitted on the source cohort alone.

These additions make cohort acquisition structure an explicit part of classifier evaluation rather than an unexamined property of the benchmark.

8. Limitations

These analyses address association with recorded metadata, not causal biological specificity. Unmeasured collection structure - referral patterns, institutional treatment protocols, sample handling absent from the released metadata - is invisible to every quantity reported here, so a low metadata-only AUC is weak evidence that a cohort is unaffected. Within Ren, omitting the recruiting hospital gives a collection-only AUC of 0.553, against 0.750 with the completed metadata.

The collection-stratified permutation conditions on a subset of the recorded metadata (Section 4.4). A null that conditions on the full matrix, including continuous columns, would require a conditional randomisation scheme that we do not develop; this is the main methodological gap of the study.

Only the collection block represents study design in the strict sense, and demographic and sample-quality associations have other interpretations, so both definitions are reported. V_D is linear and depends on coding; the nonlinear arms use two tree ensembles rather than an exhaustive set of learners; and column counts overstate effective dimension.

Residualisation results are specific to ridge at a fixed penalty on a training-fold standardised matrix. The simulation shows that, under this procedure, a loss of the size seen in CMV carries no information about contamination, but not what other residualisers would do; the random-forest, overlap-weighting and location-adjustment arms in Section 5.3 already differ materially.

The decision tree organises evidence but has not been validated against an external criterion, and no result depends on it.

The nine comparisons come from five datasets, share donors within datasets, and are not independent replications. The datasets were selected to span acquisition structures, not sampled from all eligible public cohorts (Section 9.1), so eight of nine is not a prevalence estimate. Three comparisons have fewer than 15 donors in the minority class; they are flagged, and no conclusion rests on them alone. All datasets are peripheral blood from public repositories curated under one schema, so the results do not establish cross-tissue behaviour. Matched random subsets separate the cost of restriction from the cost of using fewer donors but do not identify a technical effect, so signal retained after restriction is not shown to be biological. Geneformer pretraining overlap with the benchmark cells was not tested.

Methods

9. Materials and methods

9.1. Cohorts and the unit of analysis

Seven public datasets were used, in two groups. Table S8 is the registry. It gives, for each dataset, the identifier, the donor count reported by the source, and the donor, case and control counts actually modelled - these differ, because donors below the minimum cell count or carrying a label outside the contrast are dropped. Five entries were verified against the CELLxGENE Discover API on 2026-09-07; the two used only in Section 5 come from GEO and are marked as not API-verified.

Five datasets enter the main screen, all obtained through the CELLxGENE Discover curation API: the CD4-positive alpha-beta T-cell subset of GSE174188 (261 donors; 162 cases, 99 controls) [2], the Ren COVID-19 atlas (196 donors) [39], the Stephenson COVID-19 cohort (120 donors) [40], the COMBAT consortium (124 donors) [41] and the HIHA cytomegalovirus cohort (108 donors) [42]. Nine case-control comparisons are formed from these five, so the comparisons are not independent of one another; Table 1 gives the donor overlap.

Two further lupus datasets are used only in the downstream analyses of Section 5, not in the screen: GSE135779 (44 childhood donors; 33 cases, 11 controls) [1] and GSE285773 CD4-positive T cells (26 donors; 16 cases, 10 controls) [3], the latter as the independent transfer target.

Registry entries, dataset identifiers, donor counts and download sizes were verified against the API on 2026-09-07 and are released in `cohorts/registry.yaml`.

Dataset selection. The five screen datasets were chosen before the screen was run, to span different acquisition structures rather than to sample every eligible cohort. They are a standardised single-assay atlas expected to show weak association (HIHA), a lupus cohort with processing waves

(GSE174188), a three-site COVID-19 cohort (Stephenson), a multi-centre atlas with two sequencing chemistries (Ren), and a consortium cohort in which several phenotypes share one acquisition structure (COMBAT). Each had to provide human peripheral blood, donor identifiers, a phenotype contrast within the dataset, and at least 200 cells per retained donor.

To show how large the pool was, we counted, after the analysis, the CELLxGENE Discover datasets published by 2026-09-07 that meet the screen's dataset-level requirements (`audit/cellxgene_eligibility_audit.py`, API response retrieved 2026-09-13). A dataset qualifies if it is human, carries a blood tissue label and a single-cell suspension, has a "normal" label alongside at least one disease label, and has at least 40 donors. Of 2,226 datasets, 35 in 15 collections qualify, and the five analysed collections are among them (Table S9). Dataset-level metadata do not give donor counts per class, so some of the other ten collections may fail the per-class floors. The count of eight in nine therefore describes a purposive selection covering a third of the eligible collections; it is not a prevalence estimate.

The donor was the independent unit in every supervised analysis. Cells from one donor never crossed a training/test partition.

9.2. Donor-level tables and the metadata variables

Each cohort was streamed from its h5ad in 100,000-cell chunks and reduced to one row per donor. Expression was aggregated to log1p CPM pseudobulk. The layer used for aggregation was chosen by sniffing for integer counts rather than assuming a layer name.

Design variables were drawn from schema-guaranteed observation fields (`donor_id`, `disease`, `sex`, `development_stage`, `assay`, `tissue`) plus cohort-specific collection variables detected by pattern from the distributed object. Detection patterns cover batch, pool, run, lane, chip, site, centre, city, region, province, hospital, clinic, study, sub-study, source, dataset and donor source. Ages given as strings ("25-year-old stage", "fifth decade stage", "90 year-old and over stage") were parsed to years.

Covariates that are consequences of the disease were excluded by an explicit blocklist matching severity, outcome, comorbidity, medication, sample timing, symptom, diagnosis, stage, grade and

WHO score. Supplementary Table S1 lists, for every cohort, which columns entered the metadata matrix, in which group, and which were excluded and why.

Variables were assigned to three groups. **Sample quality**: cells per donor, mean UMI per cell, mean genes per cell, aggregate mitochondrial percentage, and their logs. **Demographic**: age in years, sex, ethnicity. **Collection**: every detected `batch__*` variable, assay and suspension type.

9.3. Residual label variance

For centred phenotype-label vector y and recorded metadata matrix $\mathbf{D}$ including an intercept, V_D $= 1 - R^2(y \sim D)$. Two versions are reported. The in-sample version uses ordinary least squares on all donors and is reported for comparison. The cross-fitted version replaces the fitted values with out-of-fold predictions from five-fold `KFold` with shuffling, seeded by the run seed, and is the version used everywhere a number is interpreted. Negative R^2 values were clipped at zero before subtraction.

9.4. The metadata matrix's own permutation null

For each design matrix, the phenotype-label vector was permuted 200 times. Each permuted label was pushed through the **same five folds** used for the observed value, so the null is a distribution of the cross-fitted statistic, not of a different one. The reported p-value is one-sided, $(1 + \#\{\text{null} \leq \text{observed}\}) / (200 + 1)$, and answers: how often does a random phenotype label leave at most as much unexplained variation as the observed one? The null mean and its 2.5th percentile are reported alongside.

The metadata-only AUC is tested the same way, with 200 permutations of the phenotype label through the identical fixed-hyperparameter pipeline (`p_design_auc` in Table S3); that is the primary test, for the reason given in Section 2.2. The in-sample V_D is released with its own in-sample null, as `V_D_insample` and `p_V_D_insample` in Table S3. The two are separate quantities and are never mixed: p(V_D) always refers to the cross-fitted pair. This null is what makes either version readable, because its location depends on the number of metadata columns relative to donors.

9.5. The fitted pipelines

Four analyses fit classifiers, and they are not one pipeline. The multi-cohort screen, the decision-tree walkthrough, the calibration simulation and the GSE174188/GSE135779 deep dive each have their own settings, and Table S10 lists all of them. That table is generated by `tools/build_hyperparameter_table.py` from the `PipelineSpec` and `ForestSpec` objects in `scripts/cohorts/pipeline_core.py` that the runners themselves import, so no setting can be described here as something other than what ran. The paragraphs below state what differs; Table S10 is the authority on every value.

Shared by the screen, the walkthrough and the calibration simulation. One classifier specification, `SCREEN_FROZEN`, is used, differing only in repeat budget. It is a balanced logistic regression (liblinear, `max_iter` 5,000) at fixed inverse regularisation 1.0 with stratified five-fold outer splits. The 4,000 highest-variance features are selected on the training donors, and a PCA to 50 components (or fewer when donors are limiting) is fitted on those donors and applied to the held-out ones. The screen runs one repeat, the walkthrough twenty. A tuned variant selects the inverse regularisation from 10^{-4} to 10^{4} in decade steps by three-fold resampling inside each outer training set, over five repeats. It is reported beside the fixed pipeline as a description; the fixed pipeline is the inferential statistic, because both permutation nulls refit it.

Metadata matrices. The metadata matrix is small and dense, so no feature filter and no PCA are applied to it. Numeric variables are median-imputed from the training donors of each fold and standardised there. Categorical variables are one-hot encoded from the levels observed in those training donors, dropping the first level; a level seen for the first time in a held-out donor is encoded as the reference level. The in-sample V_D is still defined on the whole-cohort matrix, as an in-sample statistic must be, and `n_design_feature` still reports that matrix's width.

The lupus deep dive. GSE174188 and GSE135779 predate the screen and run a wider budget on a single dataset. The classifier is a balanced logistic regression (liblinear, `max_iter` 20,000) with the same inverse-regularisation grid, selected by five-fold resampling inside each outer training set, over 20 repeated stratified five-fold donor splits with integer seeds 20260801-20260820. Out-of-fold predictions are pooled within each repeat; we report the mean AUC and the 2.5th-97.5th percentile of repeat-level AUCs. That range is a descriptive split-sensitivity range. It is not a confidence interval, and it does not treat correlated folds or repeats as independent [31].

Nonlinear metadata-only arms. Two tree ensembles rather than a wider search, because the question is whether a *low-dimensional* metadata-phenotype relationship is being missed by a linear model, not which learner maximises AUC. The screen's forest uses 500 trees, unlimited depth, minimum leaf size 2, balanced class weights; the deep dive's uses 256 trees, maximum depth 4, minimum leaf size 3. Histogram gradient boosting, in the deep dive only, uses 150 iterations, learning rate 0.05, seven terminal leaves, minimum leaf size 5. All are fitted inside every outer training fold with fixed hyperparameters and are not selected against held-out donors: a tuned nonlinear arm would not be comparable with the tuned linear one under the same permutation null.

9.6. The two permutation tests

Both permutation tests and the observed statistic run one fixed-hyperparameter pipeline, refitted from the raw donor-by-gene matrix on every permutation. Within each training fold, the 4,000 highest-variance genes are selected on the training donors. A standardiser and a PCA to 50 components (or fewer when donors are limiting) are fitted on those donors and applied to the held-out ones. A balanced logistic regression at fixed inverse regularisation 1.0 is then fitted, with a single cross-fitting repeat. Every step that touches expression is therefore inside the fold, so the reported AUC is genuinely out-of-fold and the null refits the whole pipeline rather than reusing one representation. Freezing the regularisation is still necessary: comparing a tuned observed statistic against an untuned null biases p-values downward.

Every representation step is refitted inside the fold on every permutation because the code arranges it, not because the permutation test would reveal otherwise: a label-dependent step taken before the permutation appears identically in the observed and permuted runs. The property is enforced in one place, `pipeline_core.fold_contained_auc`, which every analysis in this paper calls, and it is checked by an automated gate that fails the build if a `fit_transform` appears outside a training fold in the permutation path.

The **free** test permutes the phenotype vector without restriction; it is `p_free` in the released tables.

The **collection-stratified** test permutes it only within collection strata, preserving each stratum's observed class counts; it is `p_collection_preserving` there. A stratum is the interaction of every detected `batch__*` variable with assay, where assay varies. Table 1 names the stratum variables for every comparison, and the screen writes them into its own output (`strata_definition`) so the name and the implementation cannot drift apart. The stratified scheme is the restricted permutation of Chaibub Neto and colleagues [30]; it is applied here, not introduced here.

The conditioning set is collection variables only. Sample-quality and demographic columns are in the metadata matrix but not in the strata, so this test conditions on a subset of what is recorded. State the two nulls explicitly. The free test's null is that the phenotype label is exchangeable across all donors; the stratified test's is that it is exchangeable within each collection stratum, that is, conditional independence of expression and phenotype given those strata. Section 4.4 shows what follows: when a sample-quality variable is associated with the phenotype inside a stratum and drives expression, the conditional-independence null is false even though the generating model contains no disease effect, and the test rejects it at rates rising to 95%. Those rejections are correct rejections of a false null, not type-I errors, and they are the reason a significant stratified result does not certify disease biology. Conditioning on the full matrix would need a conditional randomisation scheme for continuous columns, which we do not attempt here.

Where a cohort records no collection variable at all, the fallback is tertiles of log cells per donor. That is a sample-quality variable, so for those comparisons the test is a sample-quality-stratified permutation and is labelled as such rather than being folded in silently.

Where every stratum is label-pure, within-stratum permutation leaves all labels unchanged and the permutation group contains only the observed labelling. The reference distribution is degenerate, and the test is reported as uninformative with p = NA. That is a statement about the permutation group, not a claim that no null distribution exists in any mathematical sense; the free permutation remains defined and is still reported.

A permutation contributes to the collection-stratified null only if at least one stratum contains more than one donor and both labels; if no permutation qualifies, the p-value is undefined and reported as such rather than as a number. Main runs used 1,000 permutations, so the smallest value it can return is 1/1001 ≈ 0.0010. The V_D and metadata-only AUC nulls use 200, so their floor is 1/201 ≈ 0.0050. Every seed used the same counts; the two floors in Table 1 are these two tests, not two different runs.

9.7. Feasibility and power checks

A comparison is reported as **not answerable with these data** on one criterion only: no collection stratum holds both classes, so the stratified permutation has no non-trivial reference distribution and no conditioned estimate can be formed. A metadata-only AUC of 1.000 is recorded alongside as corroborating structure, but it is not a second trigger. A finite-sample AUC of 1.000 states that the fitted model separated these donors completely; with 21 donors and seven one-hot columns that can happen without the metadata determining the label in the population, so it is not on its own a proof of non-identifiability. In the one comparison where both hold, the label-pure strata are the stated reason.

A comparison is reported as **underpowered**, and kept in the main ordering with a flag, if it has fewer than 40 donors or fewer than 15 in the minority class. The two gates are separated because they call for different responses: the first cannot be fixed by more careful analysis, and the second can be fixed by more donors.

Restriction has its own feasibility requirement, fixed in advance and applied uniformly: a comparison takes the restricted route only if some collection stratum holds at least five donors of each class and at least 40 donors in total. Where no stratum qualifies the restricted estimate is not computed, and the comparison is reported as needing donors under overlapping conditions rather than being given an estimate from a stratum too small to support one.

The difference between the tuned and the fixed-hyperparameter pipeline AUC is reported as a column (`frozen_minus_tuned_auc`, Table S3) and is deliberately **not** used as a feasibility rule. A gap between two different algorithms does not establish that a comparison is unanswerable. Every inferential statement in this paper is made about the fixed-hyperparameter pipeline, which is the statistic both permutation nulls are built from; the tuned AUC is descriptive. “Fixed-hyperparameter” refers to the statistical pipeline throughout, and “frozen” only to a foundation model whose weights are not updated.

9.8. Seed stability

All nine comparisons were re-run at five seeds (20260907-20260911), varying the seed for fold construction, permutation draws, and the cross-fitting split used for V_D and its own permutation null. All five per-seed outputs are released rather than summarised (Figure S7, Table S11), and every range quoted in Table 1 and in the text is the minimum and maximum over those five runs. Five master seeds control fold assignment and permutation draws throughout; the seed is read at call time in every estimator, so each seed reaches the V_D null as well as the classifier.

9.9. Simulation

Cohorts of 200 donors and 100 features were generated. A latent variable z produced a site indicator and a continuous quality proxy; the diagnosis was generated as $\rho \cdot z + \sqrt{(1-\rho^2)} \cdot e$ and then either used continuously or dichotomised. Biological and technical loading vectors overlapped partially (technical = $0.7 \cdot$ random + $0.3 \cdot$ biological, renormalised). Seven regimes were crossed with $\rho \in \{0, 0.25, 0.5, 0.75, 0.9, 0.98, 1\}$ and both label types, at 200 replicates per cell, giving 19,600 simulated cohorts. The seven regimes are:

1. **additive linear**, the base case;
2. **nonlinear design effect**: tanh plus a quadratic term in the quality proxy;
3. **heteroscedastic noise**: the standard deviation depends on the site;
4. **design-by-cell-type interaction**: two feature blocks whose mixing weight depends on the site, with the biological effect confined to one block;
5. **20% case rate, matched thresholds**: the collection boundary is moved to the same quantile as the diagnosis threshold;
6. **20% case rate, offset boundary**: the collection boundary stays at the median, so the threshold and the boundary are offset;
7. **different acquisition in the target**: the external cohort is acquired under a different design mechanism. The two 20% arms are counted separately because together they separate the effect of prevalence from that of threshold alignment (Section 4.3).

Critically, the external target reuses the **source's** biological loading vector. Regenerating it makes every transfer chance-level by construction and tests nothing; only the design mechanism differs between source and target in the shift arm.

A supplementary arm quantifies what admitting a disease-caused covariate does. A severity variable was generated as $1.2 \cdot (\mathrm{y} - \bar{y})$ + noise and added to the metadata matrix. Everything else was held fixed.

The extended simulation's cohorts are 200 donors by 100 features, so it applies no feature filter and no PCA. Its classifier settings are listed in Table S10 under `extended_simulation`. They are not derived from the screen's, because a 4,000-feature filter and a PCA to 50 would be a different reduction on a 100-column matrix. Its unadjusted, residualised and restricted arms differ from each other in exactly one operation, which is the property the calibration study below was missing.

Calibration arms. Two further studies use `pipeline_core.SIMULATION`, which *is* the screen's fixed-hyperparameter specification. Arm A places 200 donors in eight collection sites, generates a sample-quality variable that predicts the phenotype within site at strength $\in \{0, 0.3, 0.6, 1.0, 1.5, 2.0\}$, and lets expression depend on the site and on that variable only - there is no phenotype-to-expression arrow. Both permutation tests then run at 200 draws each, over 300 replicates per . Only $= 0$ measures calibration; for > 0 the stratified test's conditional-exchangeability null is false by construction, because the sample-quality variable lies outside the conditioning set, so the rejection rate there is the probability of detecting a non-disease association and is reported under that name.

Arm B generates 108 donors and 4,000 features with a real biological effect and a metadata matrix assigned independently of the phenotype, at widths 1 to 92 and in one further layout that reuses the observed level sizes of the CMV batch-by-pool crossing. Unadjusted and residualised AUCs come from the same function with one argument added, so the two arms share folds, feature filter, standardiser, PCA, classifier and scoring rule and differ only in whether the training-fold ridge fit on the metadata matrix is subtracted first. Sixty replicates per cell. The sweep is shardable across machines - each cell seeds itself from its own coordinates - and a sharded run reproduces an unsharded one.

9.10. Decision-tree walkthrough

Three cohorts were traced through the tree with all evidence recomputed, and every branch value is released in Table S7. The collection strata, the metadata matrix and the classifier are imported from the screen rather than reimplemented, so Q2 and Q4 refer to the same partition, the same columns and the same fitted pipeline; the walkthrough is `SCREEN_FROZEN` at twenty repeats. The metadata matrix is built and standardised on the training donors of each split, because ridge shrinkage depends on scale.

Restriction was evaluated in the largest stratum meeting the feasibility requirement in Section 9.7 - at least five donors of each class and at least 40 in total. Where no stratum qualifies, that is reported and the comparison takes Route C, rather than being worked around by coarsening the strata or by computing an estimate from a stratum too small to support one. Matched random subsets of identical size and case/control composition were drawn 20 times from the full cohort for comparison, which is what separates the cost of conditioning from the cost of using fewer donors.

9.11. Recovery of GSE135779 collection metadata

Batch, collection year, age, sex, race, ethnicity and clinical variables were read from the original Supplementary Table 1b, and per-library sequencing metrics from Supplementary Table 1c [1]. Study names were linked to analysis donor identifiers through the GEO title/accession map; all 44 donors mapped uniquely. The restoration script writes the donor table, batch-by-label, year-by-label and batch-by-year cross-tabulations, and SHA256 hashes of every source file. For GSE174188, per-cell `Processing_Cohort` was read from the distributed CELLxGENE object and reduced to donor fractions over four waves. The dominant wave is the maximum-fraction wave; a pure-wave donor has at least 99% of analysed cells in one wave.

9.12. Representations

Frozen Geneformer. Cell embeddings used Geneformer V2-316M at repository revision `04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5` [4]; checkpoint, configuration, token dictionary and gene-median dictionary hashes are in the methods manifest. Gene identifiers were version-trimmed and mapped to the V2 vocabulary. Within each cell the ranking value was `raw count`

`/ total count × 10,000 / Genecorpus-104M gene median`, and genes were sorted descending. Sequences used V2 start/end token IDs 2 and 3, padding ID 0 and maximum length 4096, leaving at most 4094 ranked genes. The model output was the final `last_hidden_state`; token position 0 was kept as the 1,152-dimensional cell vector. The encoder was frozen. Donor representations are coordinate-wise means of sampled cell vectors, sampled without replacement with integer seed 1, capped at 500 cells per donor for GSE135779 and 1,000 for GSE174188 and GSE285773. The embedding environment used Python 3.10, PyTorch 2.5.1 and Transformers 4.46.3.

Pseudobulk. We use pseudobulk rather than a learned latent space as the donor-level representation, so that the comparison between representations is not itself mediated by a model fitted with batch covariates [5]. Stored donor-by-gene values are `log1p(CPM)`. Inside every outer training fold, gene variance was computed on training donors, the 4,000 highest-variance genes were selected, and coordinates were standardised by training-donor mean and standard deviation. HVG pseudobulk uses these values directly. PCA pseudobulk fits up to 30 components on the training values and applies the fitted map to held-out donors. For cross-cohort transfer, trimmed gene identifiers were intersected before source-only HVG selection; feature means, variances, selected genes, scaler, PCA, classifier and regularisation were all fitted on source donors and applied unchanged to the target.

9.13. Fold-contained residualisation and its variants

Unadjusted and residualised models share outer folds, feature construction, scaling, PCA and classifier selection. Inside each outer training fold, metadata variables were encoded as in Section 9.5 and a multivariate ridge regression (`alpha` 1, intercept fitted and **not** penalised) mapped design to every raw representation coordinate. Training and held-out coordinates were replaced by observed minus predicted values, after which the identical pipeline was fitted to training residuals and applied to held-out residuals. The diagnosis was never given to the residualiser.

Three further arms were run on GSE135779. A multivariate random-forest residualiser (96 trees, depth 4, minimum leaf 3) fitted only on outer-training donors, with pseudobulk genes pre-selected by training-donor variance. A batch-location arm subtracting training-batch mean offsets from training and held-out features; because it applies no empirical-Bayes scale shrinkage we describe it as ComBat-style location adjustment rather than ComBat. And an overlap-weighting arm fitting a training-only logistic diagnosis propensity model from the full design, clipping propensities to 0.025-0.975 and training the representation classifier with overlap weights.

For the central GSE174188 comparison, residualisation loss was paired by split seed. We bootstrapped the 20 paired repeat-level losses 5,000 times and subtracted the locked matched-restriction discrepancy. These intervals quantify split sensitivity conditional on the analysed donors; they are not population confidence intervals.

To probe confound leakage with a nonlinear downstream learner [28], 100 Gaussian simulations used no biological effect, a technical effect of one, and metadata-phenotype association 0.75. Random-forest prediction was compared on raw features, globally residualised features and fold-contained residualised features.

9.14. Restriction and matched donor controls

GSE174188 was restricted to dominant wave 4 and separately to pure wave 4. GSE135779 excluded the all-case batch B1 and retained B2 to B6, each containing both labels. The complete internal pipeline was re-run after restriction. For every observed stratum, 20 donor subsets were drawn without replacement from the full cohort with identical donor count and case/control count, each using one prespecified split seed. The observed-minus-matched difference asks whether the stratum is harder than expected from reduced donor number and label composition alone.

9.15. Cell-type composition

The GSE174188 object was restricted to the same CD4-positive alpha-beta T-cell population and 261 donors as the molecular analysis. Per-donor counts were formed for `T4_naive`, `T4_em`, `T4_reg` and all other or unassigned labels. We evaluated raw closed proportions and centred-log-ratio coordinates; for CLR, 0.5 was added to each donor-category count before closure, the donor mean log abundance was subtracted and the final coordinate omitted. Each representation was evaluated with and without log total CD4-cell yield, under the same repeated donor pipeline, with processing wave predicted one against the rest.

9.16. Transfer and learned pooling

Cross-batch analyses trained on waves 2 and 3 and evaluated wave 4, then reversed. Cross-cohort transfer fitted every statistic on the source cohort; target labels were accessed only after prediction. Target AUC intervals used 5,000 case/control-stratified donor bootstrap resamples and the percentile method. Paired AUC comparisons used DeLong tests on identical target donors [23] with Benjamini-Hochberg correction inside declared method families [24].

DeepSets, gated-attention multiple-instance learning and pooling by multi-head attention operated on cap-500 frozen cell embeddings [20-22]. Cell standardisation, a whitened source PCA32 projection, network weights and hyperparameters were all source-fitted. Hidden widths were 16 or 32, weight decay 0.01 or 0.1, and Adam used learning rate 0.02 for 150 epochs; five-fold source-donor cross-validation selected the configuration, three initialisations were fitted on all source donors, and target probabilities were averaged. An ordinary donor mean in the identical PCA32 space isolates the contribution of learned pooling. The PCA32 projection used for hyperparameter selection was fitted once on all source cells before source-donor folds and never used target cells or labels. Source-internal pooling AUC is therefore not an unbiased generalisation estimate, and only independent-target and paired target-donor comparisons support the learned-pooling conclusion.

9.17. Software and reproducibility

The donor-level analyses used Python 3.11-3.13 with NumPy 2.3-2.5, pandas 2.3-3.0, scikit-learn 1.9.0, PyArrow and joblib; exact per-run versions are recorded in each output manifest. Figures were produced in R 4.6.0 with ggplot2 4.0.3 and patchwork 1.3.2. The screen, the extended simulation, the permutation calibration and the decision-tree walkthrough were run on a 16-vCPU ARM container; all other analyses were run locally. Input hashes, seeds, parameter grids, donor-level predictions, complete null distributions and output manifests accompany the analysis.

Random seeds are derived with `hashlib.blake2b` from a recorded master seed and the cell identity. Python's built-in `hash` is not used: it is salted per process for strings, so it would not reproduce across interpreter sessions. The derived seed for every cell is written into the released result tables. No script contains a machine-specific path: the figure scripts locate the project root from their own file position or from `RHEUMLENS_ROOT`, and the analysis scripts take input and output paths as arguments.

Everything downstream of the compute-heavy runs is rebuilt by one command, `bash tools/build_all.sh`: the supplementary tables, Table 1, all fifteen figures, the typeset manuscript, and two checks that fail with a non-zero exit status. The first re-derives numbers quoted in the text from the result tables and requires each to appear verbatim. The second checks the length of the abstract and author summary, that the abstract is unstructured, that figures and tables are cited in the order they are declared, and that each declared item is cited at least once. It also refuses any placeholder, retired term or withdrawn phrase, and it looks inside the rendered figures and the released table headers, where a check on the manuscript text alone cannot see them. The submission package is assembled by a further script from the same outputs, so a figure number cannot differ between the manuscript and the files.

9.18. Use of generative AI

Generative AI coding assistants were used during this work: Anthropic Claude (models Claude Opus 4.1 and Claude Opus 5) through the Claude Code command-line interface, and OpenAI Codex through its command-line interface. They were used in the following places.

- **Code review and refactoring** of the analysis scripts in `scripts/cohorts/`, `sim/` and `walkthrough/`. Every statistical definition, gate and threshold in those files was specified by the authors; the model's contribution was implementation, review of implementation, and the identification of two defects that the authors then confirmed by inspection and by rerunning the affected analyses.
- **Figure code.** All figures were drawn by R scripts written with model assistance (`figures/src/*.R`). Every number plotted was read from a released result table, and the correspondence between figure and table was checked by the authors panel by panel.
- **Manuscript language and structure**: rewriting for length and readability, reorganisation of sections, and consistency checking of cross-references and reference numbering.

No text, number, figure or citation was accepted without author verification against the underlying result tables. Cohort selection, the statistical definitions, every gate and threshold, the decision of which analyses to report, and the interpretation of results were made by the authors. No data of any kind were generated by a model: every value in this paper traces to a released result table produced by the analysis scripts. The authors take full responsibility for the content of this publication.

10. Data and code availability

All cohorts are public. The lupus datasets are at GEO accessions GSE135779, GSE174188 and GSE285773; the COVID-19, influenza and CMV cohorts were obtained through the CELLxGENE Discover curation API and their dataset identifiers are in `cohorts/registry.yaml`. The MIT-licensed repository is at https://github.com/LightChainr/rheumlens. Its releases are archived on Zenodo under the concept DOI https://doi.org/10.5281/zenodo.20813922, which resolves to the latest version; this manuscript corresponds to release v3.0.0-rc2. The versioned research object accompanying this manuscript contains the cohort registry with API-verified donor counts, the donor-level interface tables, every result table underlying Figures 2 to 8, all five per-seed outputs rather than a summary, the 19,600-cohort simulation output, analysis scripts, locked environments, `REPRODUCE.md` and `SHA256SUMS`. Public raw single-cell matrices remain at their original accessions.

11. Declarations

Author contributions. Conceptualization, H.Y. and D.L.; methodology, H.Y.; software, H.Y.; validation, H.Y. and D.Y.; formal analysis, H.Y.; data curation, H.Y.; writing - original draft preparation, H.Y.; writing - review and editing, D.Y. and D.L.; visualization, H.Y.; supervision, D.L.; project administration, D.L. All authors have read and agreed to the submitted version of the manuscript.

Funding. This research received no external funding.

Competing interests. The authors declare no competing interests.

Ethics approval. Not applicable. This study analysed only publicly available, de-identified single-cell datasets released by their original investigators under the consent and approvals obtained at each contributing site. No new human or animal data were collected.

Informed consent. Not applicable, for the reason given above.

Use of AI tools. See Section 9.18, which states what was used, where, and how the output was checked.

Acknowledgments. The authors thank the investigators of GSE135779, GSE174188 and GSE285773, and the contributors to the COMBAT, Stephenson, Ren and CMV cohorts distributed through CELLxGENE Discover, for releasing donor-level data that made this re-analysis possible.

Figure legends

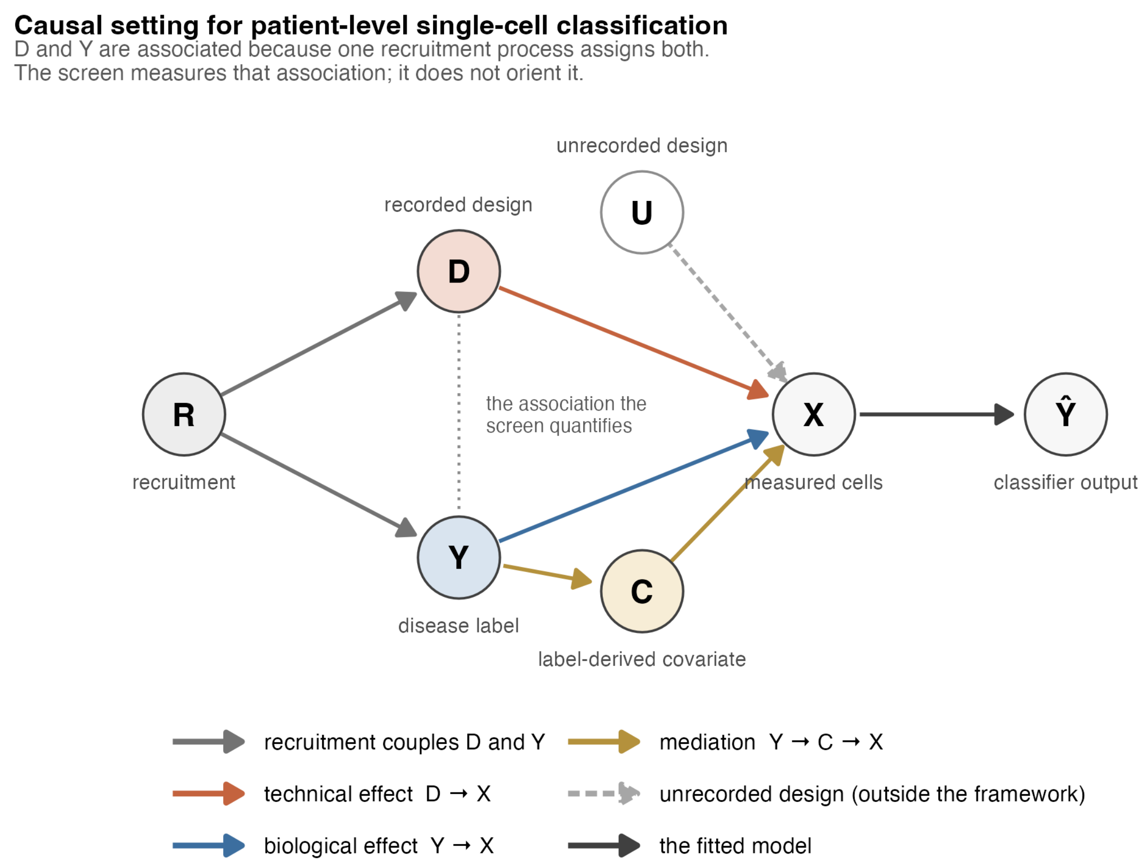

Figure 1. What is being measured. A recruitment process R places each donor in the recorded design D and is also associated with the phenotype label Y; the measured cells X are affected by D (technical) and by Y (biological); a classifier maps X to $\hat{Y}$. C denotes covariates caused by the disease - severity, medication, symptom timing - which lie on $Y \rightarrow C \rightarrow X$ and are excluded from D by blocklist. U denotes collection structure that was never recorded and which nothing in this paper can see. The dotted line marks the association these checks quantify. They measure it; they do not tell you which way it runs.

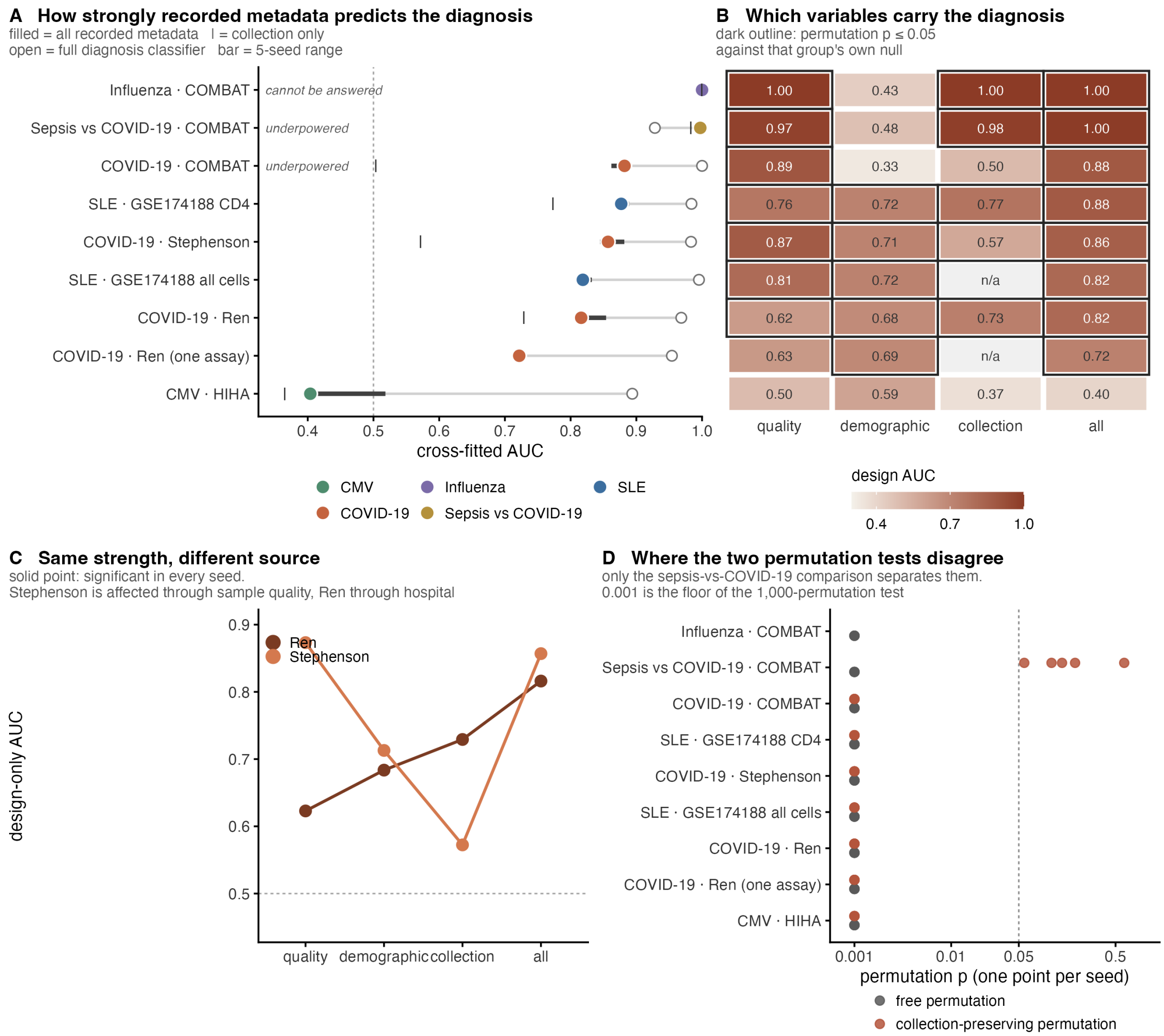

Figure 2. Nine phenotype contrasts. (A) Comparisons ordered by cross-validated metadata-only AUC (filled, coloured by disease) with the corresponding expression-classifier AUC (open) on the same axis. Bars give the range over five random seeds. Dashed line is chance. Italic labels mark the two comparisons that the feasibility check removes from or flags within the ordering. **(B)** Metadata-only AUC by variable group, rows aligned to (A). A dark outline marks $p \leq 0.05$ for the metadata-only AUC against that group's own permutation null; "n/a" marks a group absent for that comparison. **(C)** The two COVID-19 cohorts with almost equal overall metadata-only AUC are affected through different variable groups: Stephenson through sample quality, Ren through recruiting hospital. Solid points are significant. **(D)** Free and collection-stratified permutation p-values, one point per seed, for all nine comparisons. Dashed line is 0.05. Only the sepsis-versus-

COVID-19 comparison separates the two tests.

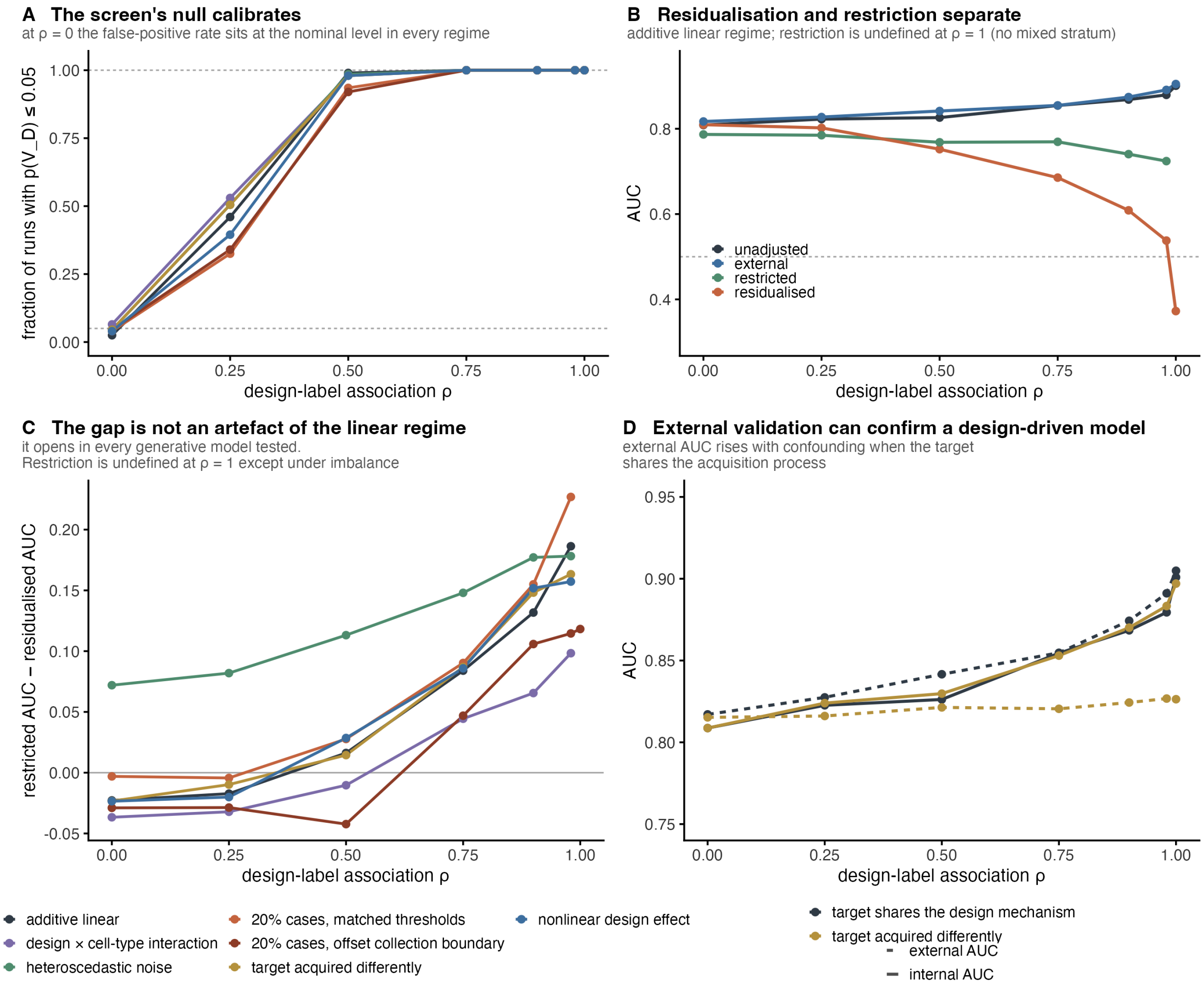

Figure 3. Simulation across seven data-generating regimes. 19,600 simulated cohorts (200 donors, 100 features): seven regimes × seven levels of metadata-phenotype association ρ × two label types × 200 replicates. **(A)** Fraction of runs with p(V_D) $\leq$ 0.05, binary labels. At $\rho = 0$ this is 0.025 to 0.065 across regimes, bracketing the nominal 0.05. **(B)** Unadjusted, external, restricted and residualised AUC against ρ in the additive linear regime. Restriction is undefined at $\rho = 1$ because no stratum holds both labels. **(C)** Restricted minus residualised AUC for all seven regimes. The gap opens in every one. It is blunted in the `imbalanced_offset` regime, where the collection boundary and the phenotype threshold sit at different quantiles, and not in `imbalanced`, where the case rate is the same 20% but the two are matched (Section 4.3). **(D)** Internal (solid) and external (dashed) AUC when the target shares the source's collection mechanism versus when

it does not. External AUC rises with confounding in the shared case.

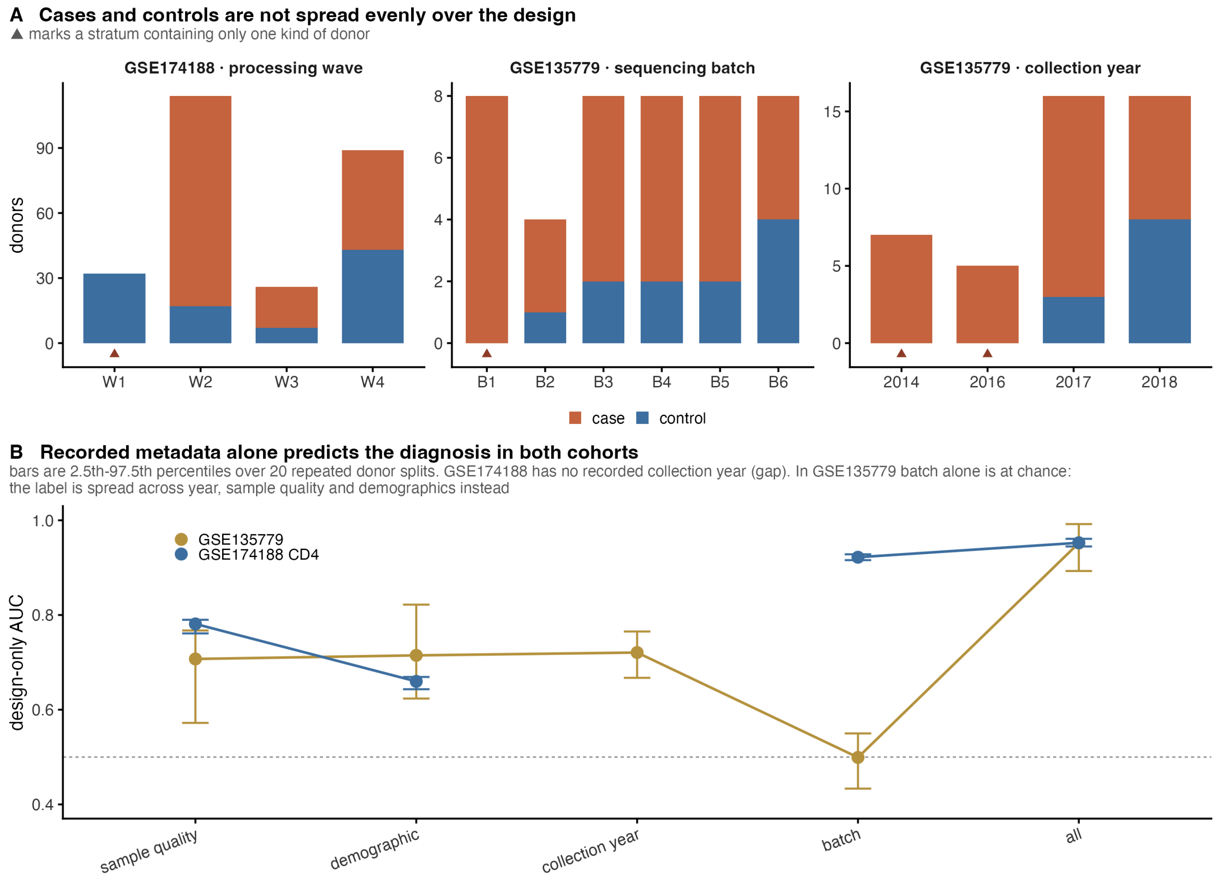

Figure 4. Design and diagnosis in the two lupus cohorts. **(A)** Donors per collection stratum, split by case and control, for processing wave in GSE174188 and for sequencing batch and collection year in GSE135779. A triangle marks a stratum holding only one kind of donor; such strata cannot be permuted within, and cannot be restricted to. **(B)** Cross-validated AUC for predicting the diagnosis from recorded metadata alone, by variable group, with 2.5th-97.5th percentiles over 20 repeated donor splits. GSE174188 has no recorded collection year, so its line is broken there. In GSE135779 batch alone sits at chance (0.499) while the other groups do not, so the association runs through collection year, sample quality and demographics rather than through batch.

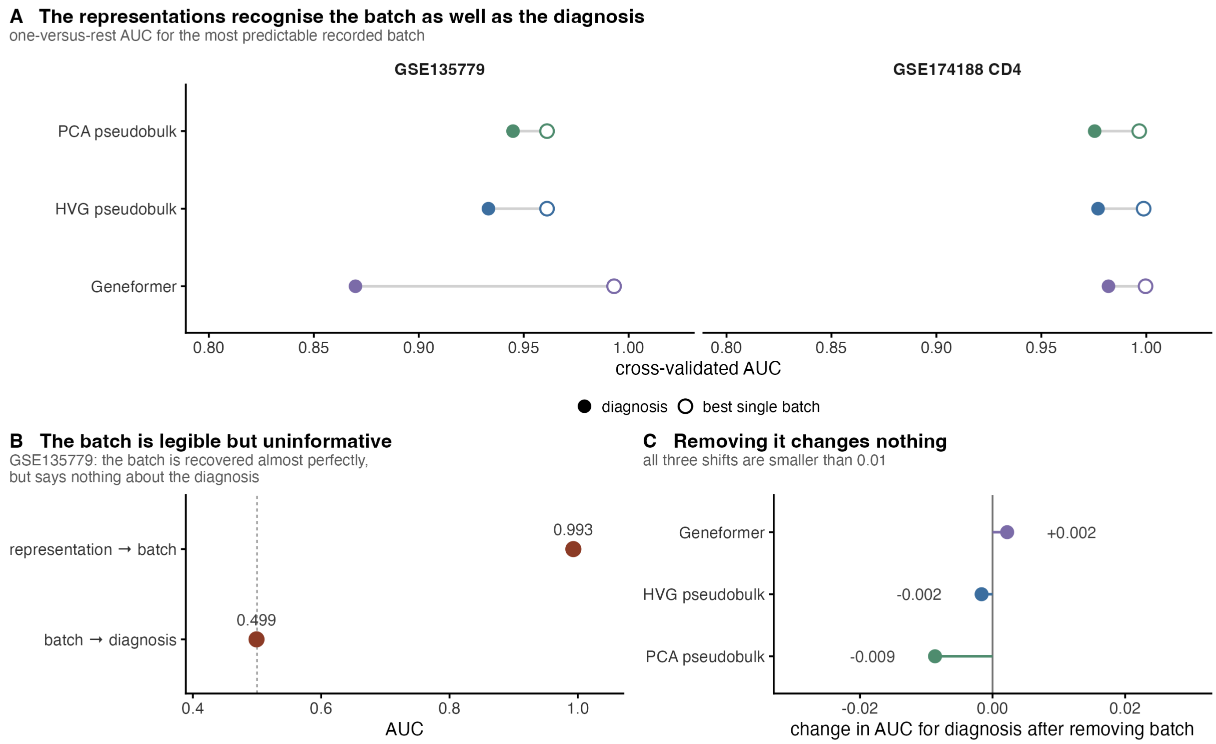

Figure 5. Containing batch information is not the same as using it. (A) For each representation, the cross-validated AUC for the diagnosis (filled) and for the most predictable single recorded batch, one against the rest (open), in both cohorts. Every representation recognises a batch about as well as it recognises the diagnosis. **(B)** GSE135779 negative control. The representation recovers the batch at AUC 0.993, but the batch predicts the diagnosis at 0.499. **(C)** Change in disease AUC after fold-contained batch residualisation in the same cohort: +0.002, −0.002 and −0.009, all inside the split-to-split spread. Batch is predictable from the representation, unpredictive of disease, and its removal changes little, so batch predictability on its own supports no conclusion about what the classifier uses.

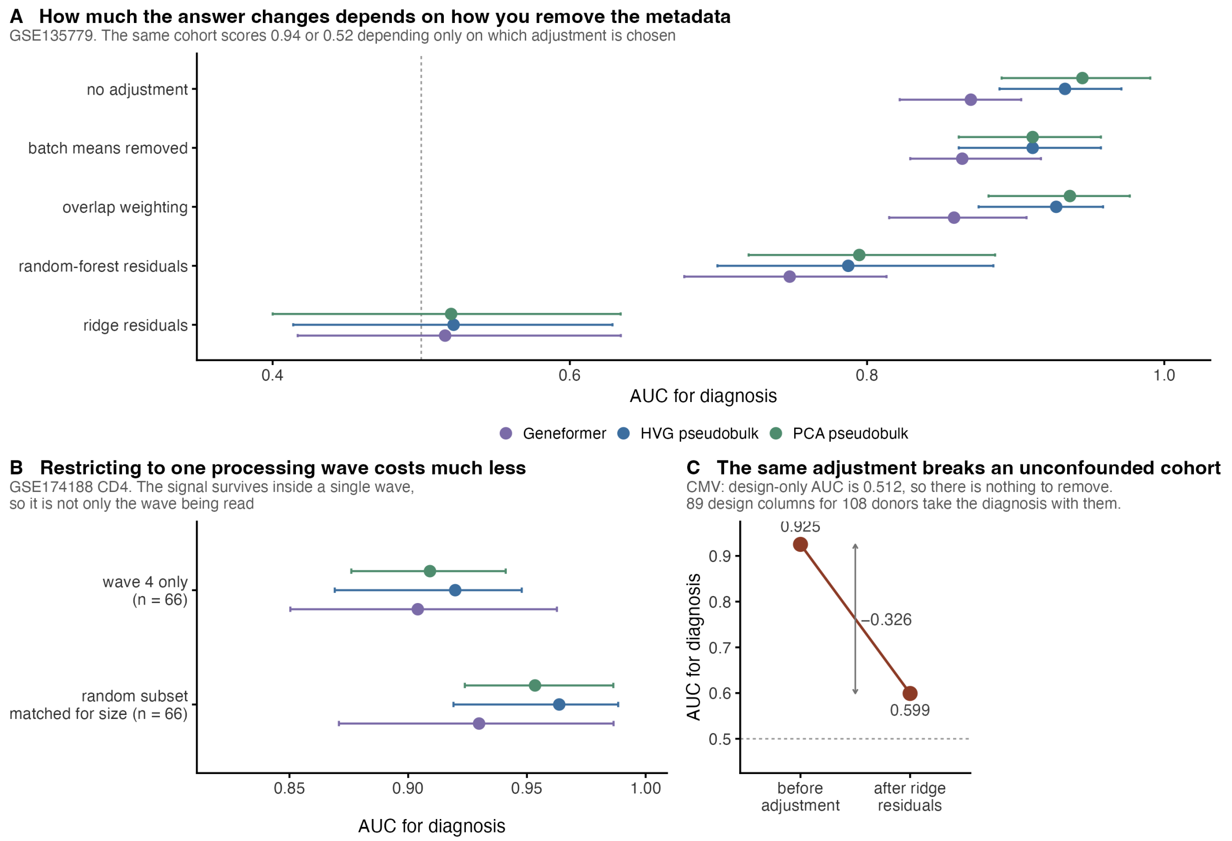

Figure 6. Residualisation and restriction give different answers, and neither is identified. (A) GSE135779 diagnosis AUC under five ways of removing the recorded metadata, for three representations, with 2.5th-97.5th percentiles over 20 repeated splits. The same cohort scores 0.94 or 0.52 depending only on which adjustment is chosen. **(B)** GSE174188 CD4, restricted to the one large near-balanced processing wave (n = 66), against size-matched random donor subsets. The separation largely survives inside a single wave. **(C)** Negative control. In the CMV comparison, where the metadata-only AUC is 0.512 and no metadata-phenotype association is detectable in any seed, the same ridge residualisation still costs 0.326 AUC. A simulated matrix of the same ragged shape, with the phenotype independent of the metadata by construction, costs 0.363 on average (central 95% 0.272-0.464; Figure S4). Residualisation loss is a function of the shape of the adjustment matrix and cannot be read as a measure of contamination.

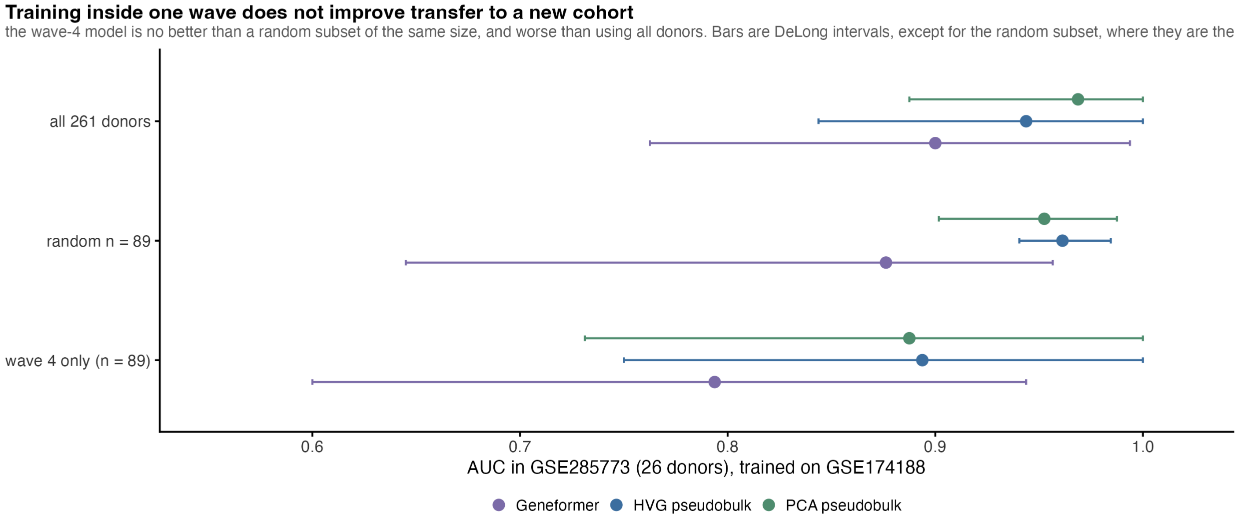

Figure 7. Restricting the training set to one processing wave does not improve transfer. AUC in GSE285773 (26 donors) for classifiers trained on GSE174188, by source set: all 261 donors, a random subset of 89, and the dominant processing wave (n = 89). The wave-restricted source is no better than a random subset of the same size and worse than using every donor. Bars are DeLong intervals, except for the random subset, where they are 2.5th-97.5th percentiles over 20 draws.

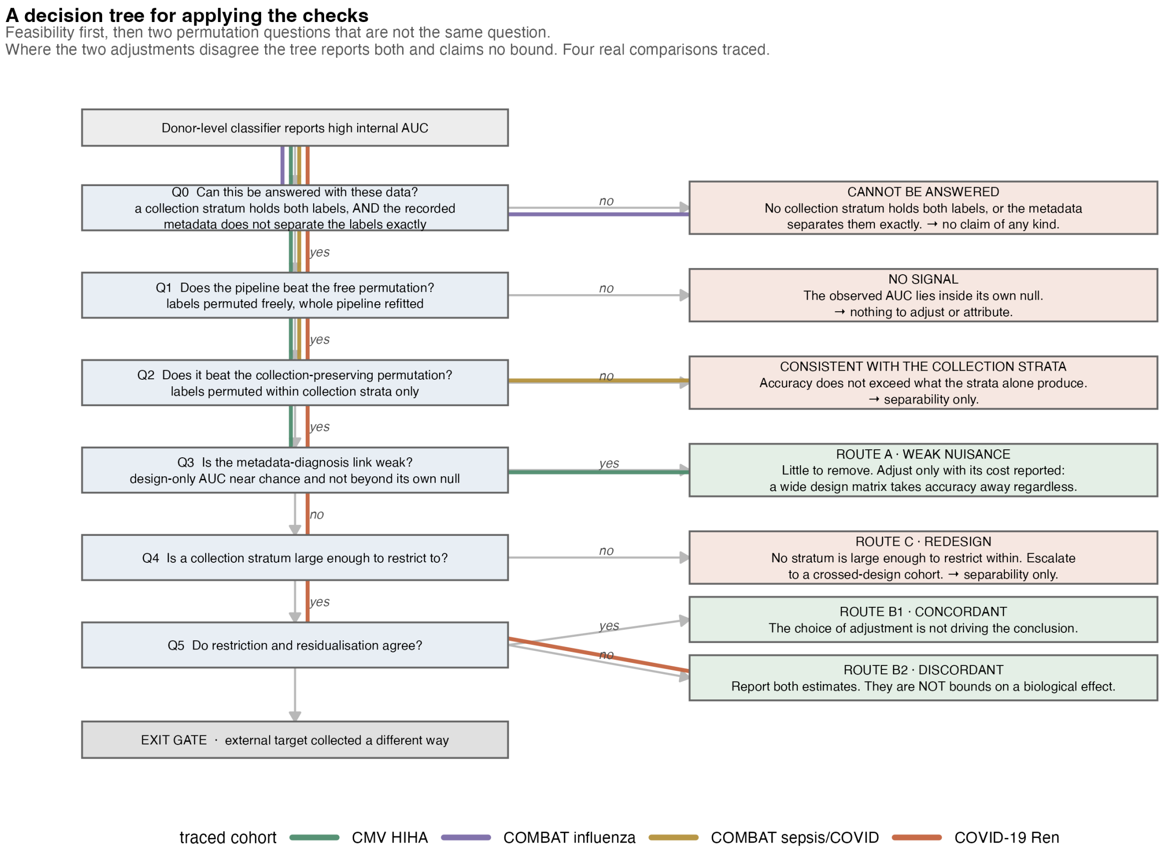

Figure 8. A decision tree for applying the checks. The feasibility question comes first (Q0): a comparison is set aside when no collection stratum holds both labels, so that no conditioned estimate can be formed. The permutation step is then split into two questions that state different nulls - the free test (Q1) and the collection-stratified test (Q2). Q3 reads the metadata-only AUC against its own permutation null rather than against a fixed cut-point; a null result there reports that no collection association was detected and leaves the unadjusted estimate standing, and does not recommend an adjustment. Q4 requires restriction to be feasible - some stratum with at least five donors of each class and at least 40 in total - so each comparison has one output; where it is not, the comparison takes Route C. Every terminal box states what can be estimated and what is still missing, not a remedy to apply. Coloured paths trace four real comparisons. The strata are the ones the collection-stratified permutation uses, not a separate definition.

Supporting information

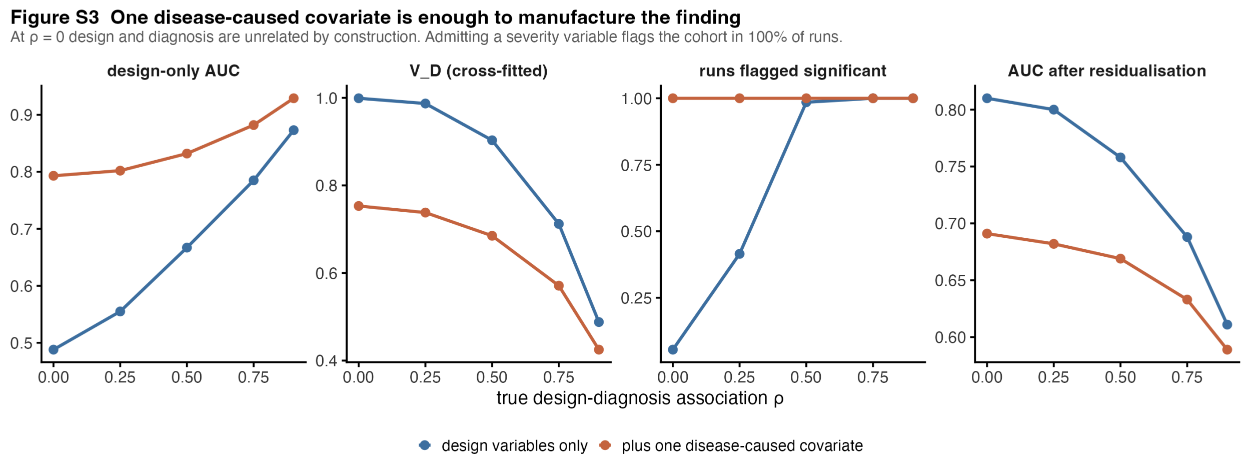

Figure S1. Admitting one disease-caused covariate manufactures the finding. Metadata-only AUC, V__D, the fraction of runs flagged significant, and residualisation loss, with and without a severity variable added to the metadata matrix, across five levels of true metadata-phenotype association. At $\rho = 0$, where metadata and phenotype are unrelated by construction, admitting the covariate raises metadata-only AUC from 0.488 to 0.793 and flags the cohort in 100% of runs against 5.5% without it.

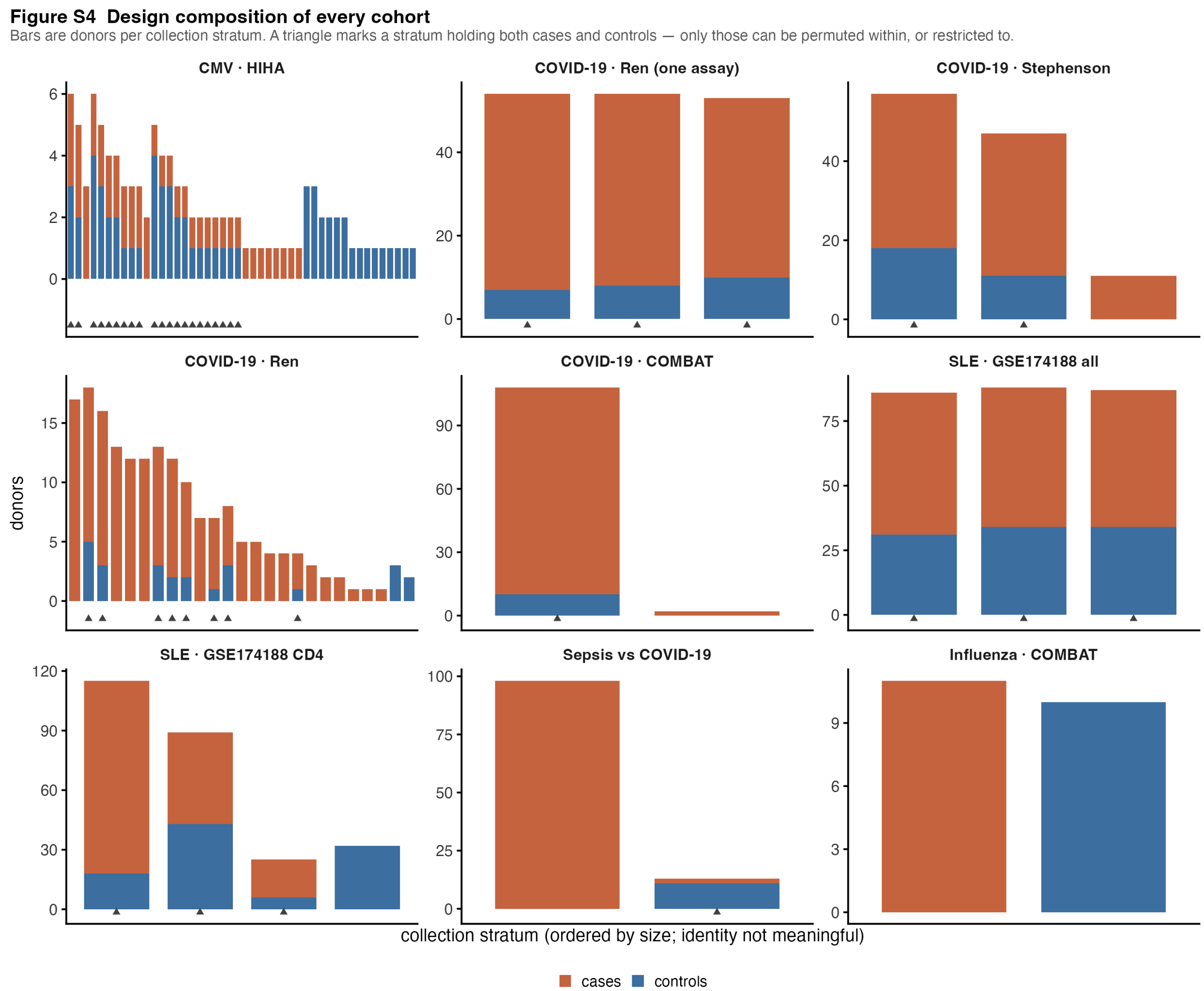

Figure S2. Design composition of each cohort. Case and control counts by collection stratum for all nine comparisons, showing which strata are label-pure and how many donors sit in mixed strata.

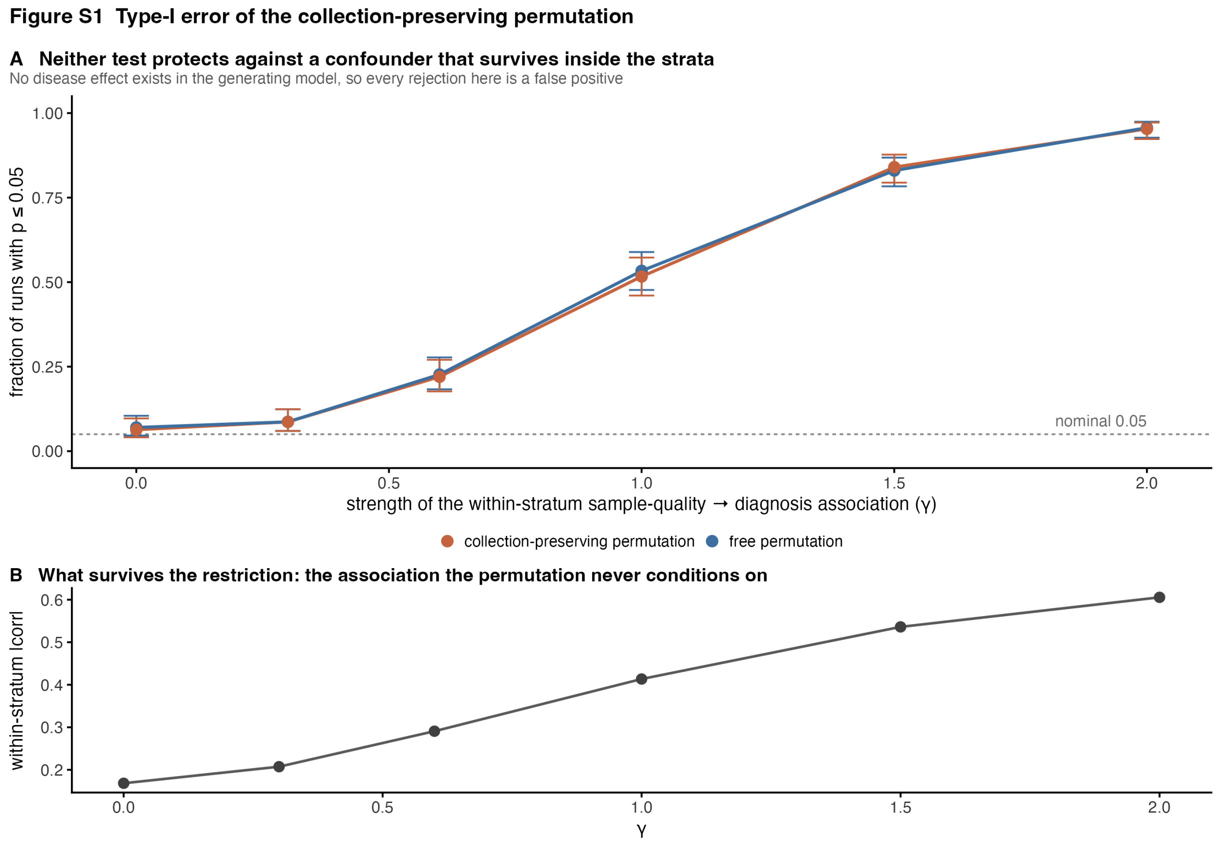

Figure S3. Rejection rate of both permutation tests under a non-disease within-stratum association. **(A)** Fraction of runs rejecting at $p \leq 0.05$ for both permutation tests, as a function of how strongly a sample-quality variable predicts the phenotype *within* a collection stratum. The generating model contains no disease effect at all. Only the leftmost point ($=$ 0) measures calibration; to its right the conditional-exchangeability null is false by construction, because the sample-quality variable lies outside the conditioning set, so the rate is the probability of detecting a non-disease association rather than a type-I error rate. **(B)** The mean absolute within-stratum correlation that conditioning on the collection strata leaves in place.

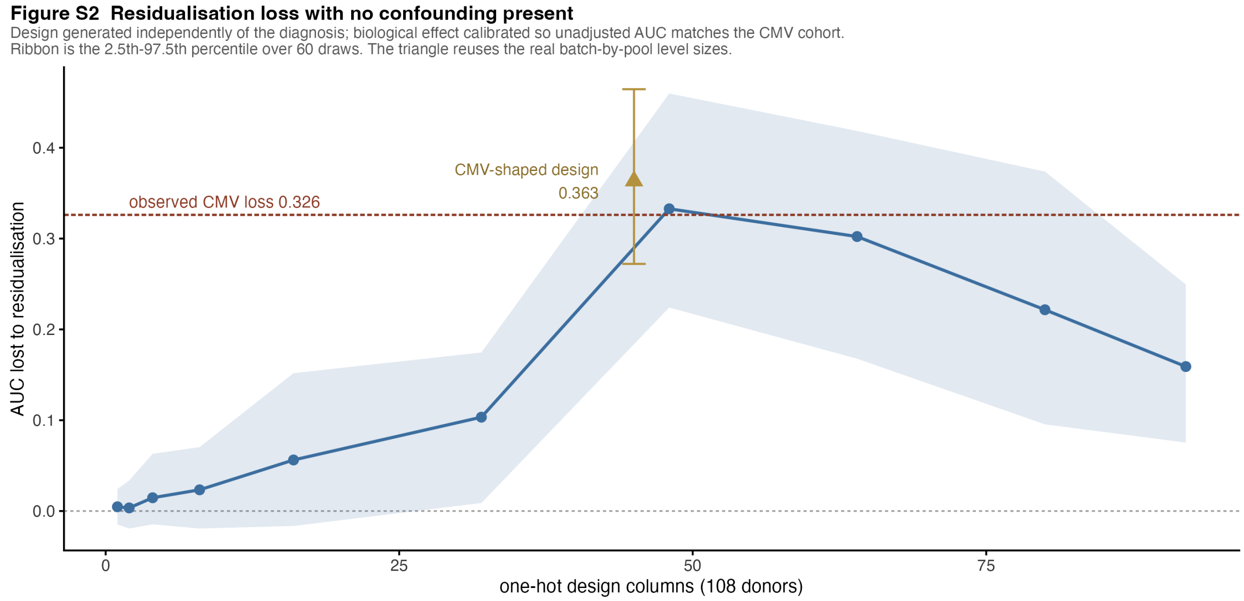

Figure S4. Residualisation loss with no metadata-phenotype association present. AUC lost to fold-contained ridge residualisation in simulated cohorts where the metadata matrix is independent of the phenotype by construction, against the width of that matrix, at 108 donors. The unadjusted and residualised arms are the same call with one argument added, so they share folds, feature filter, standardiser, PCA and classifier and differ only in the adjustment. The biological effect is calibrated so the unadjusted AUC lands near 0.90. The ribbon is the 2.5th-97.5th percentile over 60 draws. The triangle reuses the level sizes of the real CMV batch-by-pool crossing, which is far more ragged than equal blocks; it loses 0.363 on average. The dashed line marks the 0.326 observed in that comparison, which falls inside the triangle's interval.

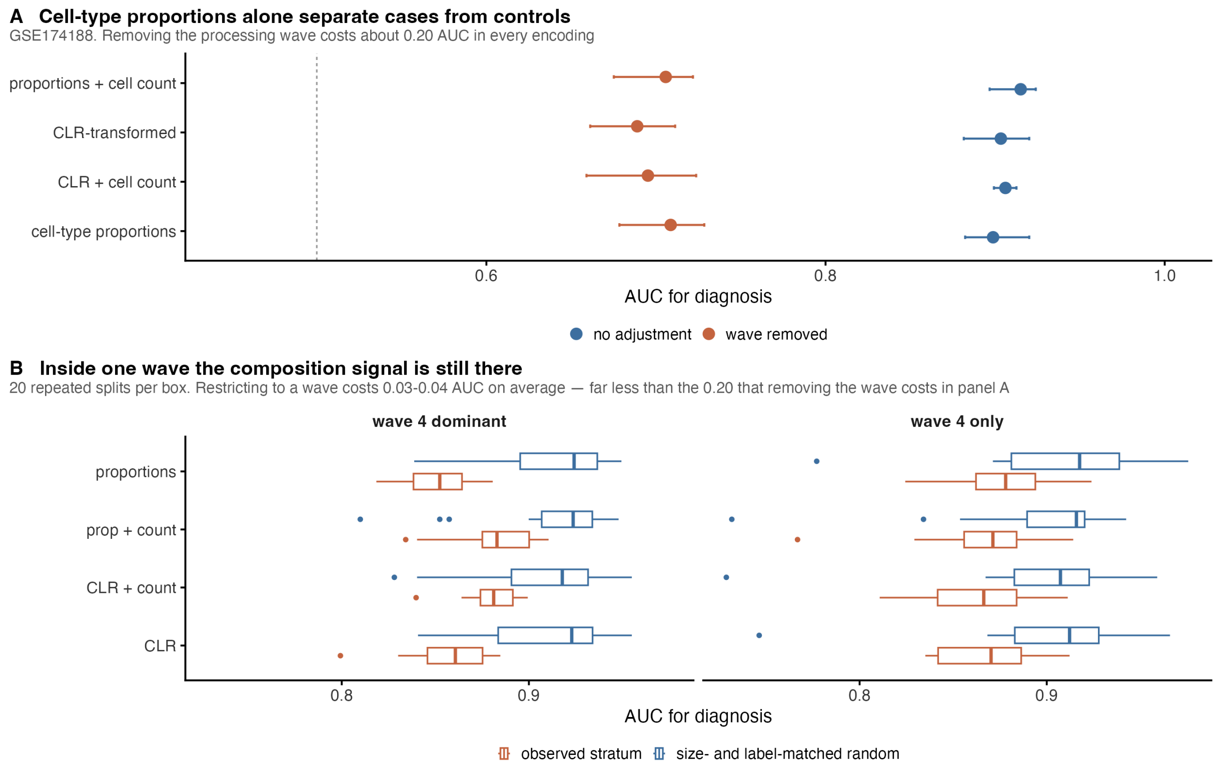

Figure S5. Cell-type composition reproduces the whole pattern. (A) Diagnosis AUC in GSE174188 from four cell-type fractions alone, under four encodings, before and after removing the processing wave. Removing the wave costs about 0.20 AUC in every encoding. **(B)** The same encodings restricted to the dominant wave and to the pure wave, each against a size- and label-matched random donor subset, 20 repeated splits per box. Restriction costs 0.03 to 0.04 AUC on average - far less than removal costs in (A). A four-number biological summary shows the same behaviour as a 1152-dimensional foundation-model embedding, which places the problem in the cohort rather than in the embedding.

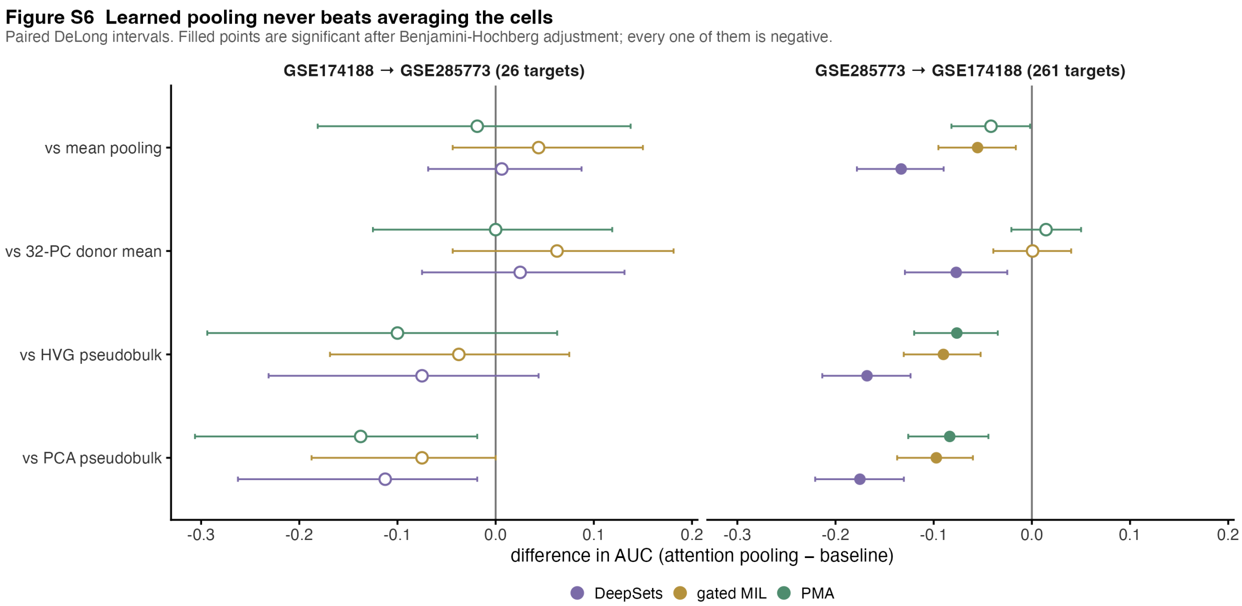

Figure S6. Learned pooling never beats averaging the cells. Paired differences in AUC between three learned pooling methods (DeepSets, gated-attention multiple-instance learning, pooling by multi-head attention) and four baselines, in both transfer directions, with paired DeLong intervals. Filled points are significant after Benjamini-Hochberg adjustment; every significant difference is negative.

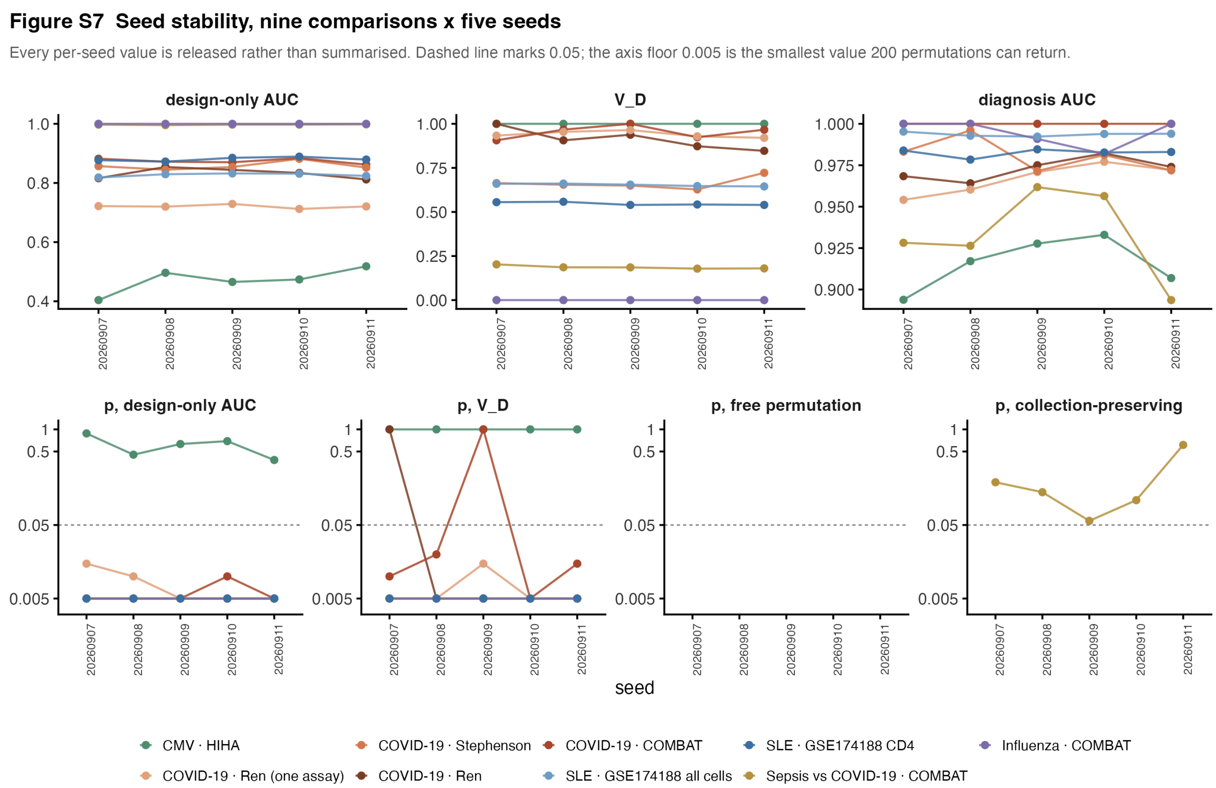

Figure S7. Seed stability. Per-seed values of metadata-only AUC and its permutation p, V_D and p(V_D), expression-classifier AUC and both permutation p-values, for all nine comparisons at five seeds. The p-value panels use a log axis; 0.005 is the smallest value 200 permutations can return.

Table S1. Design manifest, generated directly from the model-matrix construction code: for every comparison and every variable group, the raw columns consumed and the number of expanded columns produced. **Table S1b** lists every column present in a covariate file that enters no block, with the reason. **Table S1c** is a machine check that the manifest reproduces the width of every fitted model matrix (34 of 34 blocks agree). **Table S2.** Simulation summary: all seven regimes $\times$ seven ρ $\times$ two label types, plus the disease-caused-covariate arm. **Table S3.** Full screen output: every comparison $\times$ every variable group, with metadata-only AUC (linear and random forest), V_D in-sample and cross-fitted, null mean, p(V_D), expression-classifier AUC, both permutation p-values and stratum counts. **Table S4.** Metadata-only, expression-only and joint AUC for every comparison at each of five split seeds, with the two increments. All three arms use the same fixed-hyperparameter specification and the same folds; expression is reduced to principal components fitted on the training donors before the metadata columns are appended. **Table S5.** Calibration study: summary output of both arms, the permutation rejection rates of Section 4.4 and the residualisation-width study of Section 4.5. **Table S6.** Learned pooling: per-method target metrics and all paired DeLong tests, both transfer directions. **Table S7.** Decision-tree walkthrough: all evidence for every branch, three cohorts. **Table S8.** Cohort registry, generated from the registry file and the donor tables: for all seven datasets, the identifier, citation, the donor count reported by the source, the download size, the case and control labels, the donor, case and control counts actually modelled in each comparison, and how and when each entry was verified. **Table S9.** Retrospective eligibility audit of CELLxGENE Discover: every collection whose datasets meet the screen's dataset-level requirements, with donor counts, disease labels, assays and whether it was analysed. Generated by `audit/cellxgene_eligibility_audit.py`; performed after the analysis and not used for selection. **Table S10.** Every fitted pipeline in the paper and its settings, generated from the specification objects in `pipeline_core.py` that the analysis scripts import. The four analyses are separate pipelines and the table keeps them separate. **Table S11.** Seed-stability table, all five seeds, all nine comparisons.

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